

Branch-JEPA: Finite-Support Predictive Distributions for JEPA World Models

Zhi Song^{1,2}, Ximing Xing², Zhenchao Tang², Hanbo Huang², Jiehui Huang², Weilong Yan², Tianxu Lv², Minghao Yang², Zhongzheng Niu², Fan Xu², Bing He^{*2}, Lusheng Wang^{*1}, Jianhua Yao^{*2}

¹Department of Computer Science, City University of Hong Kong, Hong Kong, China

²Tencent, Shenzhen, 518057, China

Abstract

Joint-embedding predictive architectures (JEPAs) learn dynamics by predicting future observations in representation space. Yet most JEPA world models return one latent successor, even when hidden intent, partial observation, or stochastic dynamics make several futures plausible. We introduce Branch-JEPA, which replaces this point-valued transition with a context-weighted finite set of latent successors. Every branch is decoded independently, and the complete set is retained at inference. The architecture supports two complementary training regimes: specialization for recovering separated successors and full-set Energy-Score training for distributional fidelity. In a locked five-seed evaluation on the Argoverse 2 official validation split, full-set training improves trajectory Energy Score by 5.8–6.5% and probability-weighted trajectory distance by 9.3–10.4% over matched- $K=6$ assignment and transport objectives, while retaining 5.36 endpoint-deduplicated effective branches. In a parameter-exact official-validation comparison, latent branching retains 10.3% more effective modes and improves Energy Score, expected ADE, and Brier in all five paired seeds over branching only at the output decoder; every paired 95% interval excludes zero. In an OGBench graph audit, Branch-JEPA increases teleport verified-route existence to 19.2% versus 3.9% for the MDN. Its raw-support advantage also persists with 29-D state and RGB observations. Together, latent branching preserves more distinct futures, while full-set scoring improves the quality of the resulting predictive distribution.

Introduction

World models should represent the futures that remain possible given an agent’s current information. JEPAs learn such dynamics by predicting encoded targets rather than reconstructing every observation (LeCun 2022; Assran et al. 2023; Bardes et al. 2024). This abstraction has made them a natural basis for latent world models (Zhou et al. 2025a; Assran et al. 2025). Most JEPA predictors, however, return a single latent future. That is sufficient when the observed history and action determine the next state, but not when hidden intent, partial observation, or stochastic dynamics leave several futures plausible.

The limitation is structural. A point prediction can summarize an ambiguous transition, but it cannot enumerate separated outcomes. Changing the point loss changes the

summary, not its cardinality; fusing the outputs of a gated mixture has the same restriction. We call this the *point-valued support bottleneck*. It concerns what the predictor exposes, not whether its encoder has learned informative features.

Adding heads alone does not solve the problem. Heads can duplicate one another, ignore rare futures, or spread into unsupported regions. A useful multi-future predictor must both preserve distinct candidates and learn where their probability mass belongs. Multiple-choice and multimodal prediction expose the resulting trade-off: hard assignment encourages specialization but updates only the selected location, whereas joint distributional training must balance coverage and fidelity (Guzman-Rivera, Batra, and Kohli 2012; Lee et al. 2016; Rupprecht et al. 2017; Makansi et al. 2019; Narayanan et al. 2021; Seo et al. 2020).

Branch-JEPA makes multiplicity part of the JEPA transition. One encoded context produces K latent successors and context-only weights. The same decoder maps each successor independently, preserving branch identity through deployment instead of averaging the candidates back into one prediction.

How the branches are trained is a separate choice. We study two regimes for the same architecture. *Specialization* uses hard assignment to recover separated successors. *Full-set* training scores every branch and its probability mass with a proper multivariate score.

We test the architecture and training regimes separately. An exactly parameter-matched control keeps a single latent successor and branches only in the output decoder. On official validation, Branch-JEPA retains 10.3% more effective modes in this comparison and improves Energy Score, expected ADE, and Brier in all five paired seeds. Matched- K experiments on Argoverse 2 compare the two regimes’ support–fidelity trade-offs, while OGBench (Park et al. 2025), AntMaze, and RGB diagnostics test whether specialization exposes successor structure beyond motion forecasting. We evaluate fidelity, calibration, effective support, and map-aware quality rather than nominal branch count alone. Figure 1 summarizes the architecture.

This study makes three contributions:

- We introduce a branch-preserving JEPA transition whose weighted latent successors remain distinct through shared decoding; a parameter-exact control isolates its advantage

*Corresponding author.

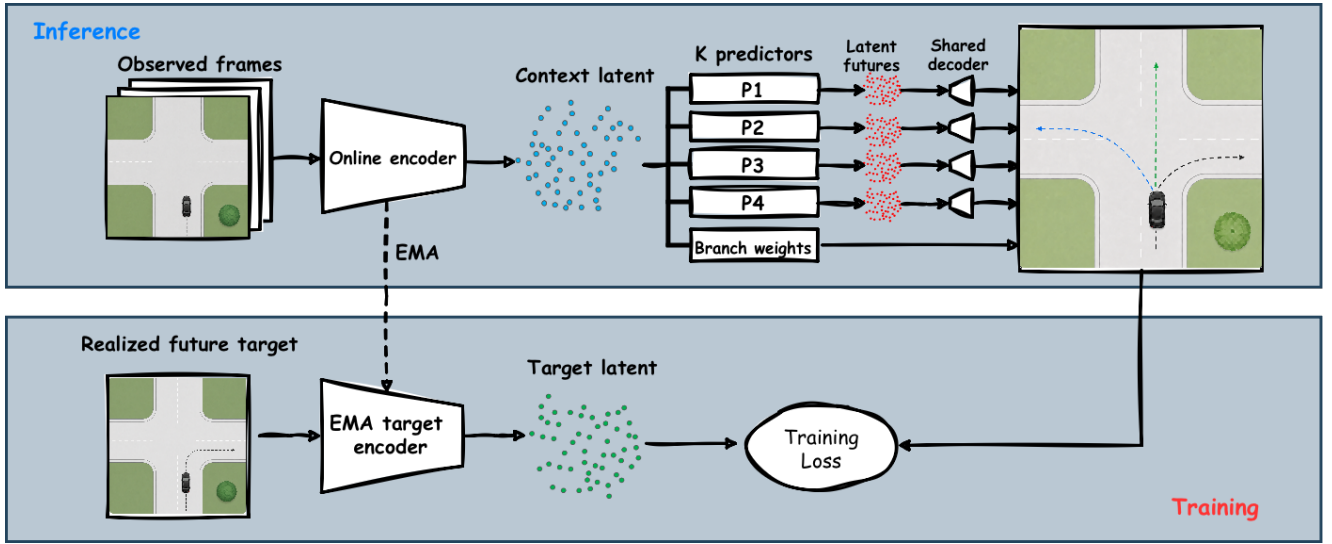


Figure 1: **Branch-JEPA architecture.** The deployable path encodes the observed context once, produces K weighted latent successors, and decodes each one independently with the same decoder; candidates are never averaged. During training, the EMA target encoder represents the realized future, which supervises the branches through either specialization (Eq. 6) or full-set scoring (Eq. 9). The target path is removed at inference, where both regimes return the complete weighted set. Four branches are drawn for clarity; road scenes and latent point clouds are schematic.

over branching only at the output decoder.

- We formulate two complementary training regimes: hard specialization for separated successors and latent/decoded full-set scoring for distributional fidelity, with a population characterization as a K -atomic energy-discrepancy approximation.
- We validate the architecture and objectives through locked matched- K AV2 experiments, frozen-anchor and official-map audits, and specialization diagnostics on OGBench, AntMaze, and RGB observations.

Throughout the paper, “Branch-JEPA” is the method name. Full-set and specialization name training objectives only, and are shown only where the objectives are compared.

Related Work

JEPA and predictive uncertainty. I-JEPA and V-JEPA established prediction in representation space (Assran et al. 2023; Bardes et al. 2024); DINO-WM and V-JEPA 2 extend latent prediction toward world modeling and planning (Zhou et al. 2025a; Assran et al. 2025; Terver et al. 2026). Probabilistic formulations use variational predictions (Huang 2026; Gögl and Yau 2026), while the concurrent UWM-JEPA preprint carries a belief through blind rollout (Radha and Goktas 2026). D-JEPA and the concurrent JEDI preprint instead use generative prediction (Chen et al. 2025; Lim et al. 2026). Branch-JEPA returns an enumerable, fixed-budget atomic set in one pass and scores that same measure on both sides of a shared decoder. M3-JEPA targets cross-modal alignment with multi-gate experts (Lei et al. 2025); its weighted-sum readout motivates our fused-output control.

Multiple hypotheses and trajectory distributions.

Multiple-choice and multiple-hypothesis learning specialize several outputs through hard assignment (Guzman-Rivera, Batra, and Kohli 2012; Lee et al. 2016; Rupprecht et al. 2017); trajectory-wise MCL applies it to dynamics models (Seo et al. 2020). Later work documents collapsed or spurious hypotheses and proposes annealing, data association, or diversity-aware sampling (Makansi et al. 2019; Narayanan et al. 2021; Yuan and Kitani 2020). Finite trajectory distributions are represented by probabilistic anchors in Multi-Path and discrete sets in CoverNet (Chai et al. 2020; Phan-Minh et al. 2020), and by learned intention queries in MTR (Shi et al. 2022). ModeSeq sequentially decodes modes with target-matched assignment (Zhou et al. 2025b). MDNs (Bishop 1994), Trajectron++ (Salzmann et al. 2020), and MotionDiffuser (Jiang et al. 2023) offer continuous or sampled alternatives. Forecast-MAE, QCNet, and Wayformer are stronger forecasting backbones (Cheng, Mei, and Liu 2023; Zhou et al. 2023; Nayakanti et al. 2023); Branch-JEPA isolates the complementary question of how a JEPA predictor should represent conditional support. Scenario-aware prediction and contingency planning (Cui et al. 2021; Chen, Ivanovic, and Pavone 2022; Chen et al. 2023) provide natural consumers of this enumerable interface. Hard assignment supplies Branch-JEPA’s specialization regime; the contribution is the branch-preserving JEPA architecture and its evaluation alongside full-set distributional training, not a new assignment rule.

Full-distribution scoring. The Energy Score is a proper multivariate score (Gneiting and Raftery 2007). DISCO Nets learn sample-based distributions (Bouchacourt, Kumar, and Nowozin 2016); SampleNet learns a finite empirical distri-

bution with Energy-Score and Sinkhorn training (Harakeh et al. 2023). Related uses include multivariate forecasting (Kan et al. 2022), trajectory evaluation (Shahroudi, Lepson, and Kull 2024), visual autoregression (Shao, Meng, and Zhou 2025), and diffusion transitions (De Bortoli et al. 2025). Branch-JEPA differs by coupling one weighted atomic measure across an EMA-target latent space and a shared decoded space, yielding both a JEPA-native distributional target and a stable atom identity from latent prediction to decoded deployment. Our matched- K comparison holds the Branch-JEPA architecture fixed while comparing full-set scoring, uniform masses, hard and soft assignment, partial optimal transport, and a development-tuned likelihood mixture.

Branch-JEPA

JEPA variables. For sample i , let s_i be the observed context, a_i an optional action, and s_i^+ the future block. The online encoder f_θ and its exponential-moving-average target copy f_ξ produce

$$\begin{aligned} h_i &= f_\theta(s_i), \quad z_i^+ = \text{sg}(\text{norm}(f_\xi(s_i^+))), \\ c_i &= [h_i, a_i]. \end{aligned} \quad (1)$$

where sg stops gradients and action-free domains set $c_i = h_i$. After each optimizer step, $\xi \leftarrow \tau\xi + (1 - \tau)\theta$. A standard predictor maps c_i to one normalized vector. Its point-valued output cannot enumerate separated conditional successors; averaging gated experts does not change that fact.

Point-valued support bottleneck. For a fixed context c , the population minimizer of squared point prediction is

$$g^*(c) = \arg \min_g \mathbb{E}[\|g - Z^+\|_2^2 | c] = \mathbb{E}[Z^+ | c]. \quad (2)$$

If predictions are constrained to the unit sphere and trained with cosine distance, the minimizer is the normalized direction of this conditional mean whenever it is nonzero. Equation 2 is the common squared/cosine special case, not a characterization of every point loss. Under a generic distance D , including the $K=1$ Energy-Score control used below, the optimizer is instead a one-point Fréchet representative $g_D^*(c) \in \arg \min_g \mathbb{E}[D(g, Z^+) | c]$. Mean or Fréchet representative, a singleton cannot enumerate separated conditional support even when the encoder retains it. A K -atomic output removes this cardinality restriction because it can place different atoms in different regions. This expressivity statement does not imply that a particular loss or optimizer will discover every region; effective support must be measured empirically.

Multi-branch predictor and shared inference contract. Branch-JEPA uses K unfused predictors g_{ϕ_k} and a context-only router r_ψ :

$$u_{ik} = \text{norm}(g_{\phi_k}(c_i)), \quad \pi_{ik} = \text{softmax}(r_\psi(c_i))_k, \quad (3)$$

which define

$$Q_{\theta, \phi, \psi}(\cdot | c_i) = \sum_{k=1}^K \pi_{ik} \delta_{u_{ik}}. \quad (4)$$

The router receives only c_i and parameterizes context-dependent event weights. At inference, Branch-JEPA returns the full set $\{(u_{ik}, \pi_{ik})\}_{k=1}^K$: alternatives remain individually addressable rather than being collapsed into a weighted average.

Decoded branches and block distance. Let $Y_i^+ = (y_{i1}^+, \dots, y_{iH}^+)$ be a future trajectory block. Latent atom u_{ik} denotes the entire block, and a shared decoder produces $\hat{Y}_{ik} = d_\omega(u_{ik})$. For decoded trajectories and $0 < \beta \leq 1$ we use

$$D_\beta(X, Y) = \left(\frac{\sum_{t=1}^H \beta^{t-1} \|x_t - y_t\|_2^2}{\sum_{t=1}^H \beta^{t-1}} \right)^{1/2}; \quad (5)$$

for normalized latents, $D_z(u, z) = \|u - z\|_2$.

Two training regimes. Branch-JEPA fixes the multi-branch architecture and its set-valued inference contract; we study two ways to train it. *Specialization training* selects the closest branch for each realized target,

$$\begin{aligned} k_i^* &= \arg \min_k D(X_{ik}, T_i), \\ \mathcal{L}_{\text{spec}}^D &= \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} D(X_{ik_i^*}, T_i) - \frac{\lambda_r}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \log \pi_{ik_i^*}. \end{aligned} \quad (6)$$

This classical hard-assignment rule drives explicit division of labor among branches; in the AV2 instantiation, the same selected branch receives latent and decoded regression updates. *Full-set training* instead treats the output as one weighted atomic distribution. Given support X_{ik} , target T_i , weights π_{ik} , and either distance, write $d_{ik}^T = D(X_{ik}, T_i)$ and $d_{ikl}^X = D(X_{ik}, X_{il})$. Its empirical Energy Score is

$$\begin{aligned} \mathcal{L}_{\text{ES}}^D &= \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \left[\sum_{k=1}^K \pi_{ik} d_{ik}^T \right. \\ &\quad \left. - \frac{1}{2} \sum_{k,l=1}^K \pi_{ik} \pi_{il} d_{ikl}^X \right]. \end{aligned} \quad (7)$$

The observation term penalizes mass far from the realized target. The pairwise term evaluates the spread of the complete weighted support; duplicated atoms receive no pairwise credit. Unlike ad hoc repulsion, the two terms form one proper distributional score. Every positive-mass atom appears in both terms, although atoms with near-zero mass and nonconvex optimization can still yield effective under-utilization. The two regimes therefore share encoders, branches, router, decoder, and prediction-time output; only their learning signal differs.

Population target. For a fixed context, let P be the true conditional law and let $\mathcal{A}_{\leq K}$ contain all probability measures with at most K atoms. On a compact strong-negative-type metric space $(\mathcal{X}, D)$ (Gneiting and Raftery 2007; Lyons 2013), define the energy discrepancy

$$\mathcal{E}_D(P, Q) = 2\mathbb{E}D(X, Y) - \mathbb{E}D(X, X') - \mathbb{E}D(Y, Y'),$$

where $X, X' \stackrel{\text{iid}}{\sim} Q$ and $Y, Y' \stackrel{\text{iid}}{\sim} P$. Define $\text{ES}_D(Q, y) = \mathbb{E}D(X, y) - \frac{1}{2}\mathbb{E}D(X, X')$. Then

$$\mathbb{E}_{Y \sim P} \text{ES}_D(Q, Y) = \frac{1}{2}\mathcal{E}_D(P, Q) + \frac{1}{2}\mathbb{E}D(Y, Y'). \quad (8)$$

Consequently, a population minimizer over $\mathcal{A}_{\leq K}$ exists and is an energy-discrepancy-optimal weighted atomic approximation to P . If $P \in \mathcal{A}_{\leq K}$, the optimal measure is uniquely P , although its slots remain non-identifiable under permutation or co-located mass splitting. Outside that well-specified case, the projection need not be unique. This statement gives the unrestricted finite-support objective a population target; it neither guarantees semantic mode recovery nor global optimization by a shared neural parameterization. Both D_z and D_β have strong negative type because they are Euclidean metrics after restriction or a positive linear rescaling. The compactness premise is automatic for the unit-sphere latent space. In decoded Euclidean space, the identity and strict identification require finite first D_β moments; existence additionally requires a compact range or an appropriate coercivity condition. The proof is in the supplement.

The joint AV2 instantiation uses $\beta = 1$ and

$$\mathcal{L}_{\text{AV2}} = \lambda_z \mathcal{L}_{\text{ES}}^{D_z}(u, z^+) + \lambda_y \mathcal{L}_{\text{ES}}^{D_1}(\hat{Y}, Y^+) + \lambda_{\text{rec}} \mathcal{L}_{\text{rec}}, \quad (9)$$

where $\mathcal{L}_{\text{rec}} = \text{SmoothL1}(d_\omega(\text{sg}(z^+)), Y^+)$ and $\lambda_z = \lambda_y = \lambda_{\text{rec}} = 1$. D_z is a dimensionless latent chord distance, D_1 is a root-mean-square trajectory distance in meters, and SmoothL1 is averaged over batch, time, and coordinate dimensions. The fixed coefficients and normalization conventions are shared by every full-set arm. The latent score updates the online encoder, predictor, and router. The decoded score also updates the shared decoder; reconstruction updates only that decoder. The target encoder moves only by EMA. For $K = 1$, the pairwise term vanishes, yielding a one-point Fréchet predictor under the same objective rather than necessarily the arithmetic conditional mean. The matched- K study compares specialization and full-set training within Branch-JEPA; the point and parameter-exact output-only controls test output cardinality and the placement of multiplicity, respectively.

Experiments

Experimental Questions and Protocol

Evidence questions. We organize the evidence by question rather than by dataset chronology: **(Q1)** what is gained by replacing a point transition with latent branches, and do they add value beyond output-only branching; **(Q2)** how do specialization and full-set training trade mode separation, fidelity, and effective support; and **(Q3)** does broader candidate support persist across trajectory, state, and RGB observations? AV2 answers Q1–Q2 with matched controls; OG-Bench tests successor support and transition fidelity, while AntMaze and RGB test raw candidate coverage.

AV2 evaluation protocol. All candidates and weights depend only on the observed scene. Although specialization uses the realized target to assign a training branch, neither regime uses that target to select, remove, or alter candidates at inference. We evaluate the complete distribution with joint-trajectory and endpoint Energy Scores. Fixed stop/straight/left/right endpoint bins convert candidate mass into event probabilities, evaluated with Brier score (Brier

1950) and expected calibration error (ECE) (Guo et al. 2017). Stop means endpoint displacement below 2 m; otherwise the polar angle of that displacement defines left at $\geq 30^\circ$, right at $\leq -30^\circ$, and straight between those thresholds. Brier is the per-scene sum over four classes, and top-label ECE uses 10 equal-width bins. Router-top-1 ADE is mean Euclidean displacement over the 60 future steps, FDE is terminal displacement, and miss rate is $1\{\text{FDE} > 2\text{ m}\}$. They evaluate $\arg \max_k \pi_k$ and therefore require no hindsight choice. Full-distribution scores evaluate the weighted conditional law, whereas router-top-1 and $K=1$ errors evaluate a single committed prediction.

Effective support and score decomposition. Nominal K is not evidence of multiplicity. We therefore report entropy-effective branch count, heads carrying at least 1% aggregate mass, pairwise diversity, and a collision-adjusted count. The last is the inverse probability that two router samples land within latent distance $\epsilon = .10$; it equals one for duplicate atoms and is not a training objective. We also decompose Energy Score into its expected observation distance and dispersion credit. Full definitions are in the supplement. Minimum-of- K errors are used only as AV2 coverage diagnostics, never for deployment-time selection.

Data and implementation. We use the official Argoverse 2 motion-forecasting data (Wilson et al. 2021): 5 s of focal-agent and neighbor history plus local map context predict a 6 s, 60-step future. Deterministic preprocessing yields 199,908 training and 24,988 validation scenes in a focal-centric frame aligned to the last observed heading. We retain the 16 nearest observed neighbors and 48 nearest lane/crosswalk polylines, each resampled to 10 points. A one-layer GRU encodes the focal track; two-layer ReLU MLPs encode flattened neighbor tracks and polylines; one four-head attention layer pools the 64 context elements with the focal embedding as query. A two-layer ReLU projection produces a 512-dimensional latent ($h = 256$). Each Branch-JEPA atom uses a 512–512–512 ReLU MLP; the $K=1$ control uses 512–3204–512 to match predictor parameters. The router is 512–256– K , and the shared decoder is 512–512–120.

The online view contains the 50 observed focal states, observed neighbors, and static map. The target view replaces the focal track with the 60-step future; its velocity is the finite difference divided by 0.1 s and its heading is the velocity direction. Future neighbors are zeroed and masked apart from one zero dummy slot used only to avoid empty attention, while the same static map is retained. Online and target encoders are separate instances of the identical architecture; the target is initialized from the online encoder and updated only by EMA. The GRU accepts the 50- and 60-step focal sequences without an input-shape adapter. Full feature definitions and preprocessing details are in the supplement.

For the primary matched- K study, we deterministically split the official training set into 189,964 fitting and 9,944 development scenes; the 24,988-scene official validation set is excluded from model, recipe, and checkpoint selection. We train seeds $\{0, \dots, 4\}$ for exactly 30 epochs without early stopping, using AdamW, learning rate 3×10^{-4} , weight decay

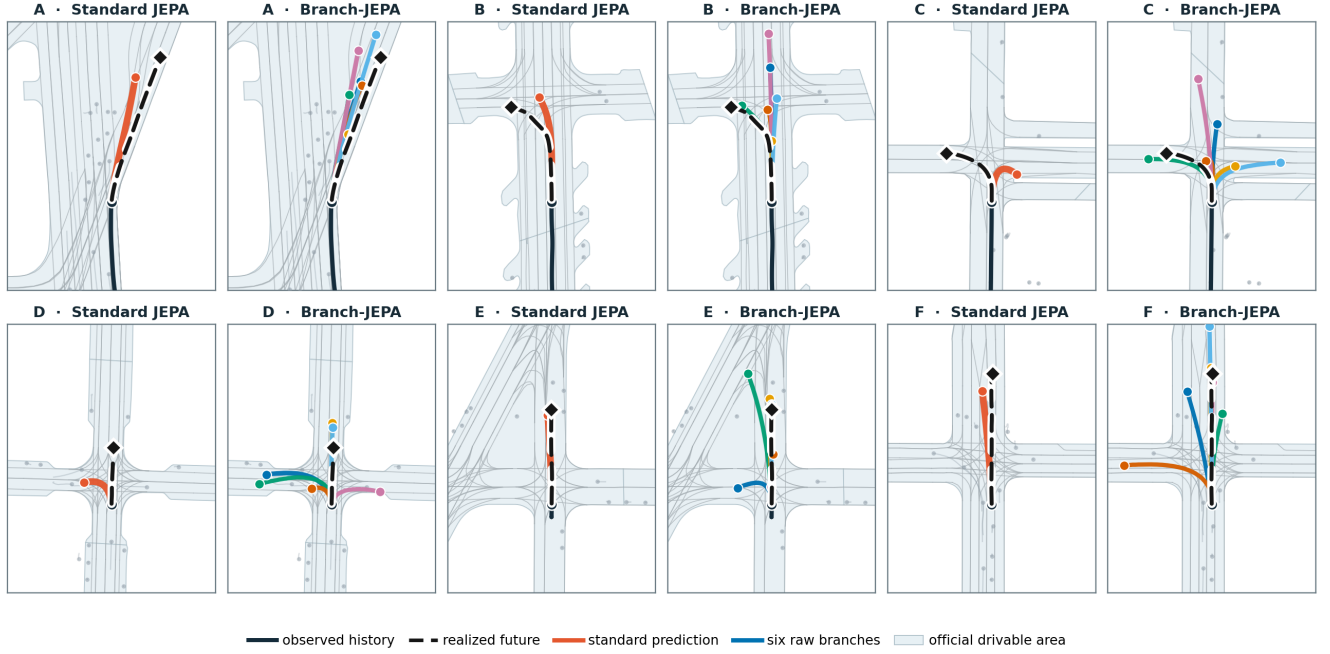


Figure 2: **Branch-JEPA adapts its finite support to contextual ambiguity on AV2.** Six target-blind cases from the frozen-anchor control are shown in a common agent-centric frame. Each adjacent pair contrasts Standard JEPA (orange, $K=1$) with all six raw Branch-JEPA trajectories (colored). The black dashed trajectory is the realized future, added only after scenario identifiers were locked; it was unavailable to either predictor and unused for case selection. In (A), branches concentrate on one broad continuation; in (B)–(F), they separate along distinct map-supported directions. Every candidate is shown; a separate target-blind official-map audit confirms that all six candidates in every displayed case remain inside the official drivable polygons, and no output is projected, snapped, filtered, or removed. Population distribution and containment results appear in Table 1 and the official-map audit.

10^{-2} , batch size 256 per GPU process, two linear-warmup epochs followed by cosine decay, mixed precision, gradient norm 5, and EMA momentum $\tau = .996$. Every arm uses 22,290 optimizer/EMA updates; within each seed, initialization, split, backbone, decoder, schedule, and checkpoint selection by minimum development trajectory Energy Score are locked. This protocol confines model and recipe selection to the train-derived development split. After locking the complete arm–seed grid, we evaluate all 35 checkpoints once on the official validation split without further tuning. Only aggregate statistics are retained; raw targets, raw predictions, and per-scene scores are not persisted. The five parameter-exact output-only checkpoints use the same development-only selection rule and are likewise evaluated once with the locked aggregate-only evaluator. Two additive $K=6$ objective controls preserve every architectural and optimization choice: soft-WTA uses one detached whole-trajectory responsibility vector at temperature .25, while partial Sinkhorn mixes hard geometry with a balanced transport plan at $\rho = .5$ and entropic regularization $\epsilon = .1$. We also steelman the MDN over a prespecified 12-configuration matrix spanning fixed, learned-global, and component-specific scales, with and without component-usage regularization. The configuration with the lowest mean development trajectory Energy

Score is selected on development ($\sigma_z = .5, \sigma_Y = 4$ m); Table 1 reports its locked official-validation result. For finite-set evaluation, the MDN exposes its six component means and mixture masses; Gaussian scales affect NLL training but are not sampled into extra candidates. The earlier frozen-anchor attribution and map audit use seeds $\{0, 1, 2\}$ and the official validation split for both selection and reporting as a separate secondary protocol; they do not select any configuration in the primary locked evaluation. The Standard JEPA baseline uses one predictor output ($K=1$) and is widened to match the six-head predictor: total parameters differ by 130 (0.0026%). Counts include the online encoder, predictor/router, and decoder, but exclude the training-only EMA target encoder. The supplement documents the anchor gates and evaluation definitions and per-seed results; the code archive contains preprocessing and writes anchor hashes, resolved arguments, and training manifests during execution.

Two Branch-JEPA Training Regimes at Matched Cardinality

Full-set versus specialization. The first and fifth rows of Table 1 are two training regimes of the same Branch-JEPA architecture. Specialization keeps all six branches active and provides the explicit mode-separation behavior used in

model	objective	ES ↓	obs. ↓	exp. ADE ↓	top-1 ADE ↓	Brier ↓	N_{eff} ↑
Branch-JEPA full-set		2.0766±.0057	3.6185±.0077	2.7941±.0059	2.4694±.0888	.09194±.00104	5.364±.023
Branch-JEPA	full-set, uniform	2.1030±.0047	3.6482±.0145	2.8214±.0108	2.7383±.4421	.09316±.00099	5.468±.050
Branch-JEPA	soft-WTA	2.2041±.0089	3.9876±.0177	3.0677±.0139	2.4858±.0223	.09352±.00054	5.132±.023
Branch-JEPA	Sinkhorn	2.2072±.0130	3.9890±.0225	3.0695±.0192	2.5143±.0117	.09388±.00047	5.150±.038
Branch-JEPA	specialization	2.2201±.0088	4.0395±.0177	3.1103±.0133	2.4932±.0125	.09360±.00057	5.125±.009
MDN	NLL, tuned	2.3827±.0839	4.0605±.1138	3.1253±.0868	2.5609±.0860	.10130±.00318	3.824±.111

Table 1: **Matched-cardinality AV2 objective matrix** (mean and sample SD across five locked seeds on the official validation split). All $K=6$ rows share the EMA-target backbone, output count, initialization protocol, training budget, and development-only checkpoint rule. Observation is the probability-weighted full-trajectory distance before the Energy Score dispersion credit; router top-1 is target-blind. Deduplication merges endpoint-connected components within 1 m. Full-set scoring gives the best mean Energy Score, observation fidelity, expected ADE, router-top-1 ADE, and Brier; all six atoms remain active. The MDN is the best mean development ES among a prespecified 12-configuration scale matrix and is evaluated here without official-split retuning. The parameter-matched $K=1$ structural control and the complete official-validation per-seed matrix, including ECE and endpoint metrics, are reported in the supplement.

our cross-domain experiments. Full-set training retains all six while lowering trajectory Energy Score by 6.46%, its observation term by 10.42%, expected ADE by 10.16%, and Brier by 1.78% relative to specialization. Against soft-WTA and partial Sinkhorn, its Energy Score improves by 5.78/5.91%, its observation term by 9.26/9.29%, and expected ADE by 8.92/8.97%. Every Energy-Score and observation comparison has the same direction in all five paired official-validation seeds. Thus the gain is not carried only by the dispersion credit: probability-weighted candidate-to-observation fidelity improves at the same time. For the direct full-set-minus-specialization comparison, the paired mean differences are $-.1435$ trajectory ES (paired- t 95% interval $[-.1574, -.1295]$) and $-.4211$ observation distance ($[-.4512, -.3909]$). Full-set training also has the best mean target-blind router-top-1 ADE among the matched- K rows.

The 12-configuration MDN sweep narrows the earlier baseline gap but does not reverse the ordering on locked official validation: full-set Branch-JEPA improves Energy Score, the observation term, expected ADE, and Brier by 12.8%, 10.9%, 10.6%, and 9.2%, respectively. It also retains 5.36 deduplicated effective branches versus 3.82 for the MDN. These official-validation results preserve the development ordering without official-split retuning.

Fixed uniform masses remain a strong control at 2.1030. Learned context weights add a further 1.25% trajectory-ES improvement, 0.82% observation improvement, and 1.31% Brier improvement, with the ES and observation direction preserved in every paired seed. The dominant gain therefore comes from learning the complete support, while context-dependent mass estimation adds a smaller, repeatable improvement. Supplementary Table S3 reports the parameter-matched $K=1$ structural reference, the complete official-validation per-seed matrix, calibration and point-error metrics, and paired statistics; Supplementary Table S10 gives the development MDN scale matrix.

Latent Branching versus Output-Only Branching

We also fix output cardinality, downstream trainable parameters (3,608,958), five seeds, 30 epochs, and the trajectory En-

ergy Score, but replace the K latent atoms by one predicted latent followed by a shared decoder trunk and K trajectory residual heads. Table 2 shows the locked official comparison. Latent branching improves Energy Score, expected ADE, Brier, and effective support in every paired seed. All four paired 95% intervals exclude zero; for Energy Score and effective modes the Branch-minus-output-only intervals are $[-.0447, -.0304]$ and $[.456, .631]$. Thus, at exactly matched downstream capacity and output count, placing multiplicity in the JEPA transition provides more effective support and better distribution quality than introducing it only after a singleton latent. Per-seed results and the exact parameter identity are in Supplementary Table S4.

Latent-objective ablation. On the train-derived development split, a five-seed same-architecture control removes all target-path gradients and trains only the decoded set. Relative to this control, the full objective reduces expected ADE from $2.7904±.0128$ to $2.7644±.0133$ and raises effective modes from $5.593±.042$ to $5.806±.010$; the paired 95% differences are $[-.0476, -.0044]$ and $[.155, .271]$. Its trajectory Energy Score is comparable (2.0560 versus 2.0577). Decoded full-set scoring therefore supplies the main task-space score, while latent supervision improves candidate fidelity and support utilization.

Predictor-Only Frozen-Anchor Control

To isolate the predictor from decoder adaptation, each seed first trains a future-view target encoder and decoder; single-branch and full-set Branch-JEPA predictors then share this identical SHA-locked, frozen anchor and receive no task-space gradient. All three anchors pass prespecified variance and effective-rank gates ($r_{\text{ent}} = 12.86\text{--}13.20$), ruling out the measured target-collapse modes. Full-set Branch-JEPA improves latent, trajectory, and endpoint Energy Scores by 26.7%, 26.3%, and 26.5% over the parameter-matched single-branch predictor, while all six atoms remain active. Crucially, collision-adjusted latent effective support is $4.26±.05$, versus 1 for the point predictor, so the gain is not explained by duplicated heads. Supplementary Tables S6–S7 report every seed and the prespecified anchor acceptance

architecture	latent→out	ES ↓	Brier ↓	$N_{\text{eff}} \uparrow$
Standard JEPA	1 → 1	2.8426±.0046	.12095±.00160	1.000
output-only branching	1 → 6	2.1142±.0046	.09554±.00043	5.254±.051
Branch-JEPA	6 → 6	2.0766±.0057	.09194±.00104	5.797±.036

Table 2: **Official-validation architecture ladder** (five locked seeds). The widened $K=1$ row is a parameter-matched structural reference. The two $K=6$ rows have exactly 3,608,958 downstream trainable parameters and isolate whether multiplicity is placed in the latent transition or only after one latent. $N_{\text{eff}} = \exp H(\pi)$. In the locked same-device batch-64 audit, end-to-end median latency is $2.89 \pm .59$ versus $2.41 \pm .22$ ms per batch for Branch-JEPA and $K=1$, respectively.

metric (%)	Standard	MDN	Branch
verified route: medium	8.3	13.4	24.7
verified route: teleport	3.5	3.9	19.2
graph precision range	62–90	14–21	49–56
29-D state raw COV-R	29.4	38.6	57.6
RGB raw COV-R	24.3	26.7	61.3

Table 3: **OGBench support and transition fidelity**. The upper block uses the state-only $K=8$ graph protocol; the lower block reports separate higher-dimensional support campaigns. Entries are seed means; protocols, SDs, and controls are in the supplement. These are offline diagnostics, not closed-loop control metrics.

gates.

Official-Map Containment Audit

In the locked primary evaluation, full-set Branch-JEPA places $99.319 \pm .041\%$ probability-weighted trajectory points inside official drivable polygons, while retaining $5.399 \pm .025$ mass-weighted map-feasible effective branches. The corresponding 95%-point-contained trajectory mass is $97.157 \pm .181\%$. The frozen-anchor audit exports every candidate and router probability without loading the realized future and restores outputs to world coordinates without projection or map matching. On 22,979 AV2 validation scenes with a vehicle or bus focal actor, full-set Branch-JEPA places $96.775 \pm .153\%$ probability mass on drivable-polygon-contained trajectories and retains $5.093 \pm .027$ mass-weighted contained effective support. Its contained endpoint diversity is $6.426 \pm .065$ m. Thus the multi-atom output remains largely inside official drivable polygons while retaining roughly five distinct contained alternatives. The full seven-metric audit and paired-scene bootstrap intervals are in the supplement; this population audit complements the target-blind qualitative cases in Figure 2.

Specialization Results Across Domains

Branch-JEPA trained with the mode-specializing objective raises macro transition precision from 14–21% to 49–56%, and verified-route existence improves to 24.7% versus 13.4% on medium and 19.2% versus 3.9% on teleport. The raw-coverage advantage also persists for 29-D AntMaze state (57.6% versus 38.6%) and RGB observations (61.3% versus 26.7%); these latter results test breadth of candidate support,

not semantic modes. Additional statistics and visualizations are in the supplement.

Interpretation

The evidence separates architecture from training regime. Specialization exposes broad, high-precision successor support on OGBench, whereas full-set training improves AV2 distribution quality and observation fidelity over hard/soft assignment and partial transport while retaining all six branches. The parameter-exact output-only control shows on official validation a pronounced effective-support advantage and systematic distribution-quality gains from placing multiplicity in the latent transition. The frozen-anchor and map audits further show that this multiplicity belongs to the predictor and survives shared decoding as roughly five distinct drivable-area-contained futures. Fixed-uniform weighting coming within 1.25% indicates that branch locations provide most of the gain, with context-dependent masses adding a smaller improvement. The cross-domain results use specialization and remain support diagnostics, not evidence for full-set training or closed-loop control.

Scope

Branch-JEPA targets the predictive-distribution interface rather than forecasting SOTA. The primary AV2 configurations are selected on development and evaluated once on official validation. Map containment does not test lane connectivity, dynamics, or collision, and the OGBench graph/set diagnostics are not closed-loop control.

Conclusion

Branch-JEPA replaces a point-valued JEPA transition with K unfused successors that survive shared decoding. On official AV2 validation, full-set scoring improves distribution quality over matched objectives. A parameter-exact five-seed control further shows that latent branching retains 10.3% more effective modes and improves Energy Score, expected ADE, and Brier over output-only branching. Specialization broadens successor support in OGBench, AntMaze, and RGB observations.

References

Assran, M.; Bardes, A.; Fan, D.; Garrido, Q.; Howes, R.; Komeili, M.; Muckley, M.; Rizvi, A.; Roberts, C.; Sinha, K.; Zholus, A.; Arnaud, S.; Gejji, A.; Martin, A.; Robert Hogan, F.; Dugas, D.; Bojanowski, P.; Khalidov, V.; Labatut, P.;

- Massa, F.; Szafraniec, M.; Krishnakumar, K.; Li, Y.; Ma, X.; Chandar, S.; Meier, F.; LeCun, Y.; Rabbat, M.; and Ballas, N. 2025. V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning. *arXiv preprint arXiv:2506.09985*.
- Assran, M.; Duval, Q.; Misra, I.; Bojanowski, P.; Vincent, P.; Rabbat, M.; LeCun, Y.; and Ballas, N. 2023. Self-supervised learning from images with a joint-embedding predictive architecture. In *CVPR*.
- Bardes, A.; Garrido, Q.; Ponce, J.; Chen, X.; Rabbat, M.; LeCun, Y.; Assran, M.; and Ballas, N. 2024. Revisiting Feature Prediction for Learning Visual Representations from Video. *Transactions on Machine Learning Research*.
- Bishop, C. M. 1994. Mixture density networks. Technical report, Aston University.
- Bouchacourt, D.; Kumar, M. P.; and Nowozin, S. 2016. DISCO Nets: DISsimilarity COefficient Networks. In *Advances in Neural Information Processing Systems*, volume 29, 352–360.
- Brier, G. W. 1950. Verification of Forecasts Expressed in Terms of Probability. *Monthly Weather Review*, 78(1): 1–3.
- Chai, Y.; Sapp, B.; Bansal, M.; and Anguelov, D. 2020. MultiPath: Multiple Probabilistic Anchor Trajectory Hypotheses for Behavior Prediction. In *Proceedings of the Conference on Robot Learning*, volume 100 of *Proceedings of Machine Learning Research*, 86–99. PMLR.
- Chen, D.; Hu, J.; Wei, X.; and Wu, E. 2025. Denoising with a Joint-Embedding Predictive Architecture. In *International Conference on Learning Representations*.
- Chen, Y.; Ivanovic, B.; and Pavone, M. 2022. ScePT: Scene-Consistent, Policy-Based Trajectory Predictions for Planning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 17103–17112.
- Chen, Y.; Karkus, P.; Ivanovic, B.; Weng, X.; and Pavone, M. 2023. Tree-Structured Policy Planning with Learned Behavior Models. In *2023 IEEE International Conference on Robotics and Automation (ICRA)*, 7902–7908.
- Cheng, J.; Mei, X.; and Liu, M. 2023. Forecast-MAE: Self-supervised Pre-training for Motion Forecasting with Masked Autoencoders. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 8679–8689.
- Cui, A.; Casas, S.; Sadat, A.; Liao, R.; and Urtasun, R. 2021. LookOut: Diverse Multi-Future Prediction and Planning for Self-Driving. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 16107–16116.
- De Bortoli, V.; Galashov, A.; Guntupalli, J. S.; Zhou, G.; Murphy, K. P.; Gretton, A.; and Doucet, A. 2025. Distributional Diffusion Models with Scoring Rules. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, 12632–12676. PMLR.
- Gneiting, T.; and Raftery, A. E. 2007. Strictly Proper Scoring Rules, Prediction, and Estimation. *Journal of the American Statistical Association*, 102(477): 359–378.
- Gögl, M.; and Yau, C. 2026. Var-JEPA: A Variational Formulation of the Joint-Embedding Predictive Architecture—Bridging Predictive and Generative Self-Supervised Learning. In *International Conference on Machine Learning*.
- Guo, C.; Pleiss, G.; Sun, Y.; and Weinberger, K. Q. 2017. On Calibration of Modern Neural Networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, 1321–1330. PMLR.
- Guzman-Rivera, A.; Batra, D.; and Kohli, P. 2012. Multiple choice learning: Learning to produce multiple structured outputs. In *NeurIPS*.
- Harakeh, A.; Hu, J. S. K.; Guan, N.; Waslander, S. L.; and Paull, L. 2023. Estimating Regression Predictive Distributions with Sample Networks. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, 7830–7838.
- Huang, Y. 2026. VJEPA: Variational joint embedding predictive architectures as probabilistic world models. In *International Conference on Machine Learning*.
- Jiang, C. M.; Cornman, A.; Park, C.; Sapp, B.; Zhou, Y.; and Anguelov, D. 2023. MotionDiffuser: Controllable Multi-Agent Motion Prediction Using Diffusion. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 9644–9653.
- Kan, K.; Aubet, F.-X.; Januschowski, T.; Park, Y.; Benidis, K.; Ruthotto, L.; and Gasthaus, J. 2022. Multivariate Quantile Function Forecaster. In *Proceedings of the 25th International Conference on Artificial Intelligence and Statistics*, volume 151, 10603–10621.
- LeCun, Y. 2022. A Path Towards Autonomous Machine Intelligence. Version 0.9.2.
- Lee, S.; Purushwalkam, S.; Cogswell, M.; Ranjan, V.; Crandall, D.; and Batra, D. 2016. Stochastic multiple choice learning for training diverse deep ensembles. In *NeurIPS*.
- Lei, H.; Cheng, X.; Qin, Q.; Wang, D.; Huang, H.; Gu, Q.; Wu, Y.; and Ji, L. 2025. M3-JEPA: Multimodal Alignment via Multi-gate MoE Based on the Joint-Embedding Predictive Architecture. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, 33902–33917. PMLR.
- Lim, J. Y.; Shah, R.; Ikram, Z.; Yu, S.; Ma, H.; Leong, T.-Y.; and Liu, D. 2026. JEDI: Joint Embedding Diffusion World Model for Online Model-Based Reinforcement Learning. *arXiv preprint arXiv:2605.13013*.
- Lyons, R. 2013. Distance Covariance in Metric Spaces. *The Annals of Probability*, 41(5): 3284–3305.
- Makansi, O.; Ilg, E.; Cicek, O.; and Brox, T. 2019. Overcoming Limitations of Mixture Density Networks: A Sampling and Fitting Framework for Multimodal Future Prediction. In *CVPR*, 7144–7153.
- Narayanan, S.; Moslemi, R.; Pittaluga, F.; Liu, B.; and Chandraker, M. 2021. Divide-and-Conquer for Lane-Aware Diverse Trajectory Prediction. In *CVPR*, 15799–15808.
- Nayakanti, N.; Al-Rfou, R.; Zhou, A.; Goel, K.; Refaat, K. S.; and Sapp, B. 2023. Wayformer: Motion Forecasting via Simple and Efficient Attention Networks. In *IEEE International Conference on Robotics and Automation*.

- Park, S.; Frans, K.; Eysenbach, B.; and Levine, S. 2025. OGBench: Benchmarking offline goal-conditioned RL. In *ICLR*.
- Phan-Minh, T.; Grigore, E. C.; Boulton, F. A.; Beijbom, O.; and Wolff, E. M. 2020. CoverNet: Multimodal Behavior Prediction Using Trajectory Sets. In *CVPR*, 14074–14083.
- Radha, S. K.; and Goktas, O. 2026. UWM-JEPA: Predictive World Models That Imagine in Belief Space. *arXiv preprint arXiv:2605.25313*.
- Rupprecht, C.; Laina, I.; DiPietro, R.; Baust, M.; Tombari, F.; Navab, N.; and Hager, G. D. 2017. Learning in an Uncertain World: Representing Ambiguity Through Multiple Hypotheses. In *ICCV*, 3591–3600.
- Salzmann, T.; Ivanovic, B.; Chakravarty, P.; and Pavone, M. 2020. Trajectron++: Dynamically-feasible trajectory forecasting with heterogeneous data. In *ECCV*.
- Seo, Y.; Lee, K.; Clavera Gilaberte, I.; Kurutach, T.; Shin, J.; and Abbeel, P. 2020. Trajectory-wise Multiple Choice Learning for Dynamics Generalization in Reinforcement Learning. In *NeurIPS*, volume 33, 12968–12979.
- Shahroudi, N.; Lepson, M.; and Kull, M. 2024. Evaluation of Trajectory Distribution Predictions with Energy Score. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, 44322–44341. PMLR.
- Shao, C.; Meng, F.; and Zhou, J. 2025. Continuous Visual Autoregressive Generation via Score Maximization. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, 54239–54253. PMLR.
- Shi, S.; Jiang, L.; Dai, D.; and Schiele, B. 2022. Motion Transformer with Global Intention Localization and Local Movement Refinement. In *Advances in Neural Information Processing Systems*, volume 35.
- Terver, B.; Yang, T.-Y.; Ponce, J.; Bardes, A.; and LeCun, Y. 2026. What Drives Success in Physical Planning with Joint-Embedding Predictive World Models? *Transactions on Machine Learning Research*.
- Wilson, B.; Qi, W.; Agarwal, T.; Lambert, J.; Singh, J.; Khandelwal, S.; Pan, B.; Kumar, R.; Hartnett, A.; Pontes, J. K.; Ramanan, D.; Carr, P.; and Hays, J. 2021. Argoverse 2: Next Generation Datasets for Self-Driving Perception and Forecasting. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*.
- Yuan, Y.; and Kitani, K. 2020. DLow: Diversifying Latent Flows for Diverse Human Motion Prediction. In *European Conference on Computer Vision*, 346–364.
- Zhou, G.; Pan, H.; LeCun, Y.; and Pinto, L. 2025a. DINO-WM: World models on pre-trained visual features enable zero-shot planning. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, 79115–79135. PMLR.
- Zhou, Z.; Wang, J.; Li, Y.-H.; and Huang, Y.-K. 2023. Query-Centric Trajectory Prediction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 17863–17873.
- Zhou, Z.; Zhou, H.; Hu, H.; Wen, Z.; Wang, J.; Li, Y.-H.; and Huang, Y.-K. 2025b. ModeSeq: Taming Sparse Multimodal Motion Prediction with Sequential Mode Modeling. In *CVPR*, 1612–1621.

Supplementary Material

Branch-JEPA: Finite-Support Predictive Distributions for JEPA World Models

Contents

S1	Submission Scope	10
S2	AV2 Data Construction	10
	Observed context	10
	Future target view	10
S3	Architecture and Optimization	10
S4	Population Target of Full-Set Training . .	11
S5	Metrics	12
S6	Per-Seed Results	13
S7	Frozen-Anchor Gates	13
S8	Official Raw-Map Audits	13
S9	Target-Blind Qualitative Protocol	13
	Extended AV2 prediction atlas	14
S10	Cross-Domain Interface Diagnostics . . .	16

S1 Submission Scope

This supplement documents one Branch-JEPA architecture under its two training regimes. AV2 compares specialization and full-set training at matched cardinality, while OGBench and AntMaze use specialization training to test explicit support recovery across representations. These experiments answer complementary architecture- and training-level questions: the cross-domain results do not compare the two objectives, and the AV2 objective comparison does not establish cross-domain planning performance. Historical ETH/UCY, SVHN, and Forecast-MAE exploratory artifacts are excluded from the formal evidence package.

S2 AV2 Data Construction

Observed context

We use the official Argoverse 2 motion-forecasting training and validation sets cited in the main paper. The cache contains 199,908 training and 24,988 validation scenes. The focal coordinate frame has its origin at the focal agent’s last observed position (time index 49) and is rotated by the negative of its last observed heading. All trajectories and map coordinates remain in meters.

The focal track contains 50 steps with features

$$[x, y, v_x, v_y, \cos h, \sin h, \text{type}],$$

where the type flag is zero. We retain the 16 closest tracks present at time index 49, using the same 50-step layout and a binary non-vehicle flag. Missing timesteps and unused neighbor slots are zero-padded and masked.

For every lane segment, we average the left and right boundaries after resampling each to 10 points. Pedestrian crossings use one resampled edge. Each polyline point contains

$$[x, y, \cos d, \sin d, \text{intersection}, \text{crosswalk}],$$

where $(\cos d, \sin d)$ is the local tangent. We retain the 48 polylines with minimum distance to the focal origin and mask unused slots. Selection uses only the observed scene and static map.

Module	Input/output	Exact construction
Focal encoder	$7 \rightarrow 256$	one-layer GRU
Neighbor encoder	$350 \rightarrow 256$	Linear–512–ReLU–Linear
Polyline encoder	$60 \rightarrow 256$	Linear–512–ReLU–Linear
Context pooling	$256 \rightarrow 256$	one 4-head attention layer
Latent projection	$512 \rightarrow 512$	Linear–512–ReLU–Linear
Each atom predictor	$512 \rightarrow 512$	Linear–512–ReLU–Linear
Router	$512 \rightarrow K$	Linear–256–ReLU–Linear
Shared decoder	$512 \rightarrow 120$	Linear–512–ReLU–Linear

Table S1: Exact architecture used by the joint and frozen-anchor AV2 experiments. The attention query is the focal GRU state; keys and values are the 16 neighbor plus 48 polyline embeddings.

Future target view

The prediction target is the focal 60-step future trajectory $Y^+ \in \mathbb{R}^{60 \times 2}$. For the JEPA target encoder, we convert it to a 60-step focal feature sequence with the same seven channels as the observed focal track. Velocity is the forward position difference divided by the AV2 interval 0.1 s; the last velocity repeats the preceding value. Heading cosine and sine are the normalized velocity direction, and the type flag is zero. Future neighbors are unavailable and are therefore all zero with a zero validity mask. The static map tensor and mask are copied from the observed context. An all-empty-attention guard leaves one zero-valued dummy neighbor slot unmasked; this contains no future information.

The online and target encoders are distinct instances of the same network. A one-layer GRU accepts the 50-step online focal view or 60-step target focal view directly; no learned length-specific adapter is used. The target output is L2-normalized and stop-gradient. In the joint model the target parameters are initialized from the online encoder and updated after every optimizer step with EMA momentum .996. The predictor and router never receive the future view at inference. The online context embedding itself is not L2-normalized before the predictor or router; normalization is applied to the target latent and to each predicted atom.

S3 Architecture and Optimization

The parameter-matched Standard JEPA uses a 512–3204–512 ReLU MLP, widening its single predictor to the parameter budget of the six-atom predictor and router. In the joint experiment the Standard-JEPA and Branch-JEPA models contain 5,074,428 and 5,074,558 trainable parameters. In frozen-anchor Stage B they contain 3,284,612 and 3,284,742 trainable parameters; the shared target encoder and decoder are frozen and excluded from those counts. Joint counts likewise exclude the training-only EMA target encoder.

The output-only branching control keeps one 512–512–512 latent predictor and moves cardinality to the task-space readout. A context-only router predicts six masses;

a shared 512–2180 decoder trunk produces a base trajectory, and six residual heads produce the six complete 60-step trajectories. Reconstruction of the EMA target latent updates only the shared base path, whereas the complete trajectory Energy Score updates the trunk, residual heads, router, online encoder, and singleton latent predictor. The Branch-JEPA predictor/router/shared-decoder and this output-only predictor/router/base-residual decoder each contain exactly 3,608,958 downstream trainable parameters (difference zero). All 396 functional residual-scale parameters receive nonzero gradients; no inactive padding parameters are used for matching. Both models emit six target-blind trajectories and use the same split, seeds, 30-epoch budget, optimizer schedule, and development-Energy check-point rule.

All six primary $K=6$ rows emit six latent candidates through a shared backbone. Branch-JEPA with learned-mass full-set training uses the two Energy Scores in the main paper. Its uniform-mass variant fixes every mass to $1/6$ and gives the router zero gradient and no prediction-time role. Branch-JEPA with specialization training selects one winner by complete 60-step trajectory distance, applies latent and trajectory regression to that same branch, and trains the router with its assignment label. Both Branch-JEPA regimes otherwise use the same multi-branch architecture and complete-set inference. The soft-WTA control computes one detached responsibility vector $\text{softmax}(-D_1/\tau)$ from each complete trajectory block at $\tau = .25$ and reuses it for latent geometry, decoded geometry, and router supervision. The partial-Sinkhorn control mixes the hard geometry plan with a balanced entropic transport plan at $\rho = .5$ and $\epsilon = .1$; its router retains the hard whole-trajectory label. Both controls are target blind at inference.

The MDN arm minimizes mixture NLL in latent and trajectory space. A prespecified 12-configuration matrix (Table S2) contains nine fixed scale pairs $\sigma_z \in \{.1, .25, .5\}$ and $\sigma_Y \in \{1, 2, 4\}$ m, learned-global scales, component-specific diagonal scales, and component-specific scales with a .01 usage-KL term. The main table reports the lowest mean development Energy Score, the fixed pair $\sigma_z = .5, \sigma_Y = 4$ m. This is a baseline-favorable development selection rather than an independent test estimate. For common evaluation we read out all six component means and predicted mixture masses without density sampling.

The primary same- K study deterministically partitions the 199,908 official training scenes into 189,964 fitting and 9,944 development scenes. For seeds $\{0, \dots, 4\}$, every $K=6$ arm shares split membership, initialization seed, data order, architecture, and schedule. Each arm uses AdamW with learning rate 3×10^{-4} and weight decay 10^{-2} , batch size 256 on one GPU, mixed precision, gradient-norm clipping at 5, two linear-warmup epochs followed by cosine decay, and exactly 30 epochs. Every run completes 22,290 optimizer steps and one EMA update per step; there is no early stopping. Among the 30 checkpoints, minimum train-derived-development trajectory Energy Score selects one checkpoint. The official validation set is not used by these jobs for training, recipe selection, or checkpoint selection. Selected epochs for seeds 0–4 are 29/30/28/29/30 (learned mass),

MDN scale family	σ_z	σ_Y (m)	usage KL	ES-traj. ↓	Brier ↓	ECE ↓	exp $H(\pi)$ ↑
component diagonal	.25	2	0	3.2942	.19082	.08763	5.10
component diagonal	.25	2	.01	3.2703	.18434	.08218	5.14
fixed isotropic	.10	1	0	3.5584	.15998	.07446	3.80
fixed isotropic	.10	2	0	3.2013	.14046	.06303	3.62
fixed isotropic	.10	4	0	2.8083	.11179	.03440	3.27
fixed isotropic	.25	1	0	3.2290	.15509	.06246	3.55
fixed isotropic	.25	2	0	3.2371	.15633	.07227	3.51
fixed isotropic	.25	4	0	2.7683	.12620	.05739	3.03
fixed isotropic	.50	1	0	3.1703	.16809	.05855	4.11
fixed isotropic	.50	2	0	3.1849	.14886	.05951	3.77
fixed isotropic	.50	4	0	2.3710	.10567	.02854	3.84
learned global	.25 init.	2 init.	0	3.4750	.12854	.02789	4.30

Table S2: **Complete five-seed MDN steelman matrix.** Entries are means on the same train-derived development split. The fixed $(\sigma_z, \sigma_Y) = (.5, 4 \text{ m})$ row has the lowest mean trajectory Energy Score and is the MDN row used in the main paper. Because the row is selected on this development matrix, it is deliberately favorable to the baseline and is not an independent test estimate.

29/28/30/30/27 (uniform mass), 29/29/30/30/30 (hard assignment), and 29/28/30/30/30 (soft-WTA), 29/30/30/30/30 (partial Sinkhorn), and 29/29/30/29/28 (development-tuned MDN). After this grid was locked, all 30 $K=6$ and five $K=1$ checkpoints were evaluated in one pass over the 24,988 official-validation scenes. The evaluator retained exactly three aggregate JSON artifacts—a run summary, the five-seed table, and a completion manifest—and persisted no raw validation target, raw prediction, or per-scene metric. All aggregate files are SHA-256 bound to the fixed checkpoint grid and evaluator source.

The older frozen-anchor study uses seeds $\{0, 1, 2\}$, batch size 512, at most 30 epochs, and patience 8. It uses the official validation split for both checkpoint selection and reporting and is therefore retained as exploratory secondary evidence rather than part of the same- K development protocol. The original same- K jobs use one internal cluster host with four NVIDIA H20 GPUs, with one training process bound to each GPU. The additive objective controls and MDN scale matrix use packed eight-V100 jobs with one process per GPU and the same optimizer-step budget. Every job runs in one frozen container image providing the Python 3.11/CUDA 12.4 runtime; the resolved training configuration is included in the code supplement.

S4 Population Target of Full-Set Training

We give the complete statement behind the full-set score in the main paper. Let $(\mathcal{X}, D)$ be a compact strong-negative-type metric space, and let P, Q be Borel probability measures on $\mathcal{X}$. For $X, X' \stackrel{\text{iid}}{\sim} Q$ and $Y, Y' \stackrel{\text{iid}}{\sim} P$, define

$$\text{ES}_D(Q, Y) = \mathbb{E}D(X, Y) - \frac{1}{2}\mathbb{E}D(X, X'),$$

$$\mathcal{E}_D(P, Q) = 2\mathbb{E}D(X, Y) - \mathbb{E}D(X, X') - \mathbb{E}D(Y, Y').$$

Here $\mathcal{E}_D$ denotes the non-square-rooted energy discrepancy. Taking expectation of the first display over $Y \sim P$ and adding and subtracting $\frac{1}{2}\mathbb{E}D(Y, Y')$ gives

$$\mathbb{E}_{Y \sim P} \text{ES}_D(Q, Y) = \frac{1}{2}\mathcal{E}_D(P, Q) + \frac{1}{2}\mathbb{E}D(Y, Y').$$

Protocol	Target representation	Trainable modules	Training loss	Validation selection
Joint / full-set	stop-grad EMA encoder	online encoder, tor/router, decoder	latent ES + trajectory ES + target reconstruction	trajectory ES
Joint / specialization	stop-grad EMA encoder	online encoder, tor/router, decoder	assigned-branch latent/trajectory regression + router CE + target reconstruction	trajectory ES
Anchor Stage A	directly trained future-view encoder	future-view encoder, decoder	reconstruction + variance floor + rank floor	decoder ADE among gate-passing epochs
Anchor Stage B	within-seed frozen Stage-A encoder	context encoder, tor/router	latent ES only	decoded trajectory ES

Table S3: Gradient scopes. Decoded trajectories in frozen-anchor Stage B are validation-only and provide no training gradient. Within a seed, the single-branch and full-set Branch-JEPA arms load the same SHA-locked Stage-A checkpoint.

The second term is constant in Q . Strong negative type gives $\mathcal{E}_D(P, Q) \geq 0$, with equality if and only if $P = Q$.

Let

$$\mathcal{A}_{\leq K} = \left\{ \sum_{k=1}^K \pi_k \delta_{x_k} : (\pi_1, \dots, \pi_K) \in \Delta_K, x_k \in \mathcal{X} \right\},$$

where zero weights and co-located atoms allow support smaller than K . At the measure level, this class is the continuous image of the compact set $\Delta_K \times \mathcal{X}^K$. Because D is continuous and bounded on the compact space, $Q \mapsto \mathcal{E}_D(P, Q)$ is continuous, so a minimizer over $\mathcal{A}_{\leq K}$ exists. The identity above shows that its minimizers are exactly the energy-discrepancy-optimal weighted atomic approximations to P . If $P \in \mathcal{A}_{\leq K}$, the minimizing probability measure is uniquely P ; slot parameters remain non-identifiable under permutation, zero-weight slots, and redistribution of mass among atoms at the same location.

When $P \notin \mathcal{A}_{\leq K}$, the best projection need not be unique because $\mathcal{A}_{\leq K}$ is nonconvex. For example, under $D(x, y) = |x - y|$, $P = (\delta_{-1} + \delta_1)/2$, and $K = 1$, every δ_x with $x \in [-1, 1]$ is optimal. The result is pointwise in a fixed context; integrating over contexts yields the corresponding conditional risk. A shared neural model only searches its realizable subfamily and nonconvex optimization need not reach the population optimum. In this paper, D_z is a restricted Euclidean metric and, for $\beta > 0$, D_β is Euclidean distance after a positive diagonal linear transformation, so both have strong negative type. The unit-sphere latent space is compact. On the noncompact decoded Euclidean space, the identity and strict identification remain valid for distributions with finite first D_β moment, whereas existence of an optimal atomic approximation additionally requires compactness or coercivity. The result would not support the same identification claim if these metrics were replaced by squared Euclidean distance.

S5 Metrics

Full-set scores. Trajectory Energy Score flattens each complete 60-step trajectory, divides it by $\sqrt{60}$, and uses Euclidean distance; this is the root-mean-square displacement of the whole block. Endpoint Energy Score uses Euclidean endpoint distance. All atom locations and probabilities enter the observation and pairwise terms. Router-top-1 selects

$\arg \max_k \pi_k(c)$ without access to the target; ties select the first stored index, so the fixed-uniform arm deterministically reports head 0 as top-1. Its ADE is the mean Euclidean displacement over 60 future steps, FDE is terminal displacement, and MR is $\mathbf{1}\{\text{FDE} > 2\text{m}\}$; ADE, FDE, and both decoded-space Energy Scores are reported in meters. Energy Score and expected ADE/FDE evaluate all atoms with their probability mass, whereas router-top-1 evaluates one target-blind commitment. A singleton can have lower point error while giving worse distributional score and no alternative support, so we report both views without treating either as a proxy for the other.

Router utilization. For scene i , $H(\pi_i) = -\sum_k \pi_{ik} \log \pi_{ik}$. We compute $\exp H(\pi_i)$ per scene and then average over validation scenes. In contrast, an “active head” has at least 1% of probability mass after aggregating router mass over the complete validation set.

Maneuver events. An endpoint with displacement below 2 m is *stop*. Otherwise its focal-frame angle is left at $\geq 30^\circ$, right at $\leq -30^\circ$, and straight between those thresholds. Candidate probabilities are summed within the four bins. Brier is the per-scene sum of four squared probability errors. ECE uses the maximum event probability and the correctness of its event, aggregated over 10 equal-width confidence bins:

$$\text{ECE} = \sum_{b=1}^{10} \frac{n_b}{n} |\text{acc}(b) - \text{conf}(b)|.$$

Collision-adjusted latent support. The reported N_{eff}^{10} is not a greedy clustering count. For normalized latents it is

$$N_{\text{eff}}^{10} = \left[\sum_{k,l} \pi_k \pi_l \mathbf{1}\{\|u_k - u_l\|_2 \leq 0.10\} \right]^{-1}.$$

It equals one when all probability is carried by mutually duplicated atoms, discounts near-duplicate pairs, and is invariant to atom ordering. We compute it per validation scene and then average over scenes.

Endpoint-deduplicated support. For the main official-validation matrix, we connect candidate endpoints whose Euclidean distance is at most 1 m, form the resulting connected components, sum router mass within each component, and report the exponentiated entropy of the component

masses. This quantity discounts endpoint-near-duplicate trajectories while preserving their total probability mass.

S6 Per-Seed Results

Tables S4 and S7 list every reported seed rather than only aggregate means. No seed or checkpoint was removed after inspecting these metrics.

Table S4 includes the three prespecified paired full-set-minus-specialization comparisons; each has the same direction in all five seeds. Router-top-1 ADE also has the best matched- K mean for full-set training; complete paired intervals for every retained metric and comparator accompany the released aggregate table.

For latent minus output-only branching, paired mean differences are $-.03758$ trajectory ES (paired- t 95% interval $[-.04475, -.03041]$), $-.03691$ m expected ADE ($[-.04834, -.02547]$), $-.003598$ Brier ($[-.005228, -.001969]$), and $+.5431$ effective modes ($[+.4556, +.6306]$). All four have the same direction in all five seeds. The top-1 ADE difference is $+.0224$ m with interval $[-.0987, +.1435]$ and is not used as evidence of an advantage.

S7 Frozen-Anchor Gates

Stage A trains a future-view encoder and normalized-latent decoder with

$$\begin{aligned}\mathcal{L}_A &= \text{SmoothL1}(d(\bar{z}), Y^+) + \mathcal{L}_{\text{var}} + .25 \mathcal{L}_{\text{rank}}, \\ \mathcal{L}_{\text{var}} &= \frac{1}{d_z} \sum_j \left[.02 - \sqrt{\text{Var}_{\mathcal{B}}[\bar{z}_j] + 10^{-4}} \right]_+, \\ \mathcal{L}_{\text{rank}} &= \frac{[8 - r_{\text{PR}}]_+}{8}, \quad r_{\text{PR}} = \frac{\text{tr}(C)^2}{\|C\|_F^2}.\end{aligned}\quad (\text{S1})$$

where $\bar{z}$ is the normalized target latent and C its minibatch covariance. Latent normalization and covariance penalties are computed in FP32. The rank-floor regularizer therefore uses the minibatch participation ratio r_{PR} . The checkpoint gate instead uses the full-validation covariance eigenvalues λ_j and entropy effective rank

$$q_j = \frac{\lambda_j}{\sum_{\ell} \lambda_{\ell}}, \quad r_{\text{ent}} = \exp \left(- \sum_j q_j \log q_j \right).$$

These are distinct summaries; r_{PR} is the exponentiated order-2 Rényi entropy and is a conservative lower bound on r_{ent} for a common covariance spectrum. A checkpoint is eligible only if decoder ADE/FDE are at most $2/4$ m, mean normalized dimension standard deviation is at least $.005$, the fraction below 10^{-3} is at most $.90$, and entropy effective rank is at least 8 . Among eligible epochs, minimum validation decoder ADE selects the checkpoint; the thresholds were fixed before the formal three-seed run.

S8 Official Raw-Map Audits

The locked primary evaluator also runs a prespecified official-polygon summary on the official validation split. Across the five Branch-JEPA seeds, probability-weighted

candidate-point containment is $99.319 \pm .041\%$, the probability mass of trajectories with at least 95% contained center points is $97.157 \pm .181\%$, and the mass-weighted feasible effective support is $5.399 \pm .025$. These are mean $\pm$ sample SD across seed-level aggregates and establish center-point containment, not lane connectivity or dynamic safety.

Separately, for the frozen-anchor attribution audit, we export all six trajectories and router probabilities from every checkpoint, restore them to world coordinates, and evaluate the 22,979 validation scenes whose focal actor is VEHICLE or BUS. The realized future is not loaded. Predictions are neither projected nor map-matched. Geometry is the union of official AV2 drivable-area polygons with 0.1 m boundary tolerance; $22,977/22,979$ observed focal origins fall inside this union, providing a coordinate-transform sanity check.

A candidate is feasible when at least 95% of its 60 center points are inside. Feasible candidates are connected into duplicate components when their trajectory RMS distance is at most 0.5 m. Component probabilities are summed. Unique support counts components with probability at least $.01$. Feasible entropy support is the exponentiated entropy of feasible component mass after renormalization; it is set to zero for a scene with no feasible candidate, and the reported value averages over all scenes. This zero convention explains why the point-model mean can be below one. The mass-weighted version multiplies this count by total feasible probability mass. The diversity quantities below also retain the original router mass. For feasibility indicators f_k , trajectory RMS distance D_{kl}^{traj} , and endpoint distance D_{kl}^{end} , we report

$$\begin{aligned}\text{Div}_{\text{traj}} &= \sum_{k,l} \pi_k \pi_l f_k f_l D_{kl}^{\text{traj}}, \\ \text{Div}_{\text{end}} &= \sum_{k,l} \pi_k \pi_l f_k f_l D_{kl}^{\text{end}}.\end{aligned}$$

Thus they decrease when probability leaves the feasible set; they are not conditionally renormalized diversity scores.

The audit establishes substantial polygon-contained support, not non-inferior compliance: full-set Branch-JEPA's drivable-trajectory mass is 1.39 percentage points lower. Center-point containment also does not test directed lane connectivity, actor footprint, collision, dynamics, traffic rules, or closed-loop safety.

S9 Target-Blind Qualitative Protocol

Figure 2 in the main paper is a mechanism illustration, not a quantitative validity test. Its frozen prediction-time screen accesses observed focal and neighbor histories, the 48 cached lane/crosswalk polylines, official static-map polygons, scenario identifiers, and model outputs only. It never loads the realized future, a future-selected candidate index, or an error to the target. Predictions are rendered unchanged: there is no map snapping, projection, candidate deletion, or probability renormalization.

For the parent gallery, a candidate point is within the cached-map proxy when it is at most 3 m from one of the cached polylines. A context passes when its probability-weighted outside-point fraction is at most $.10$ and at least $.80$

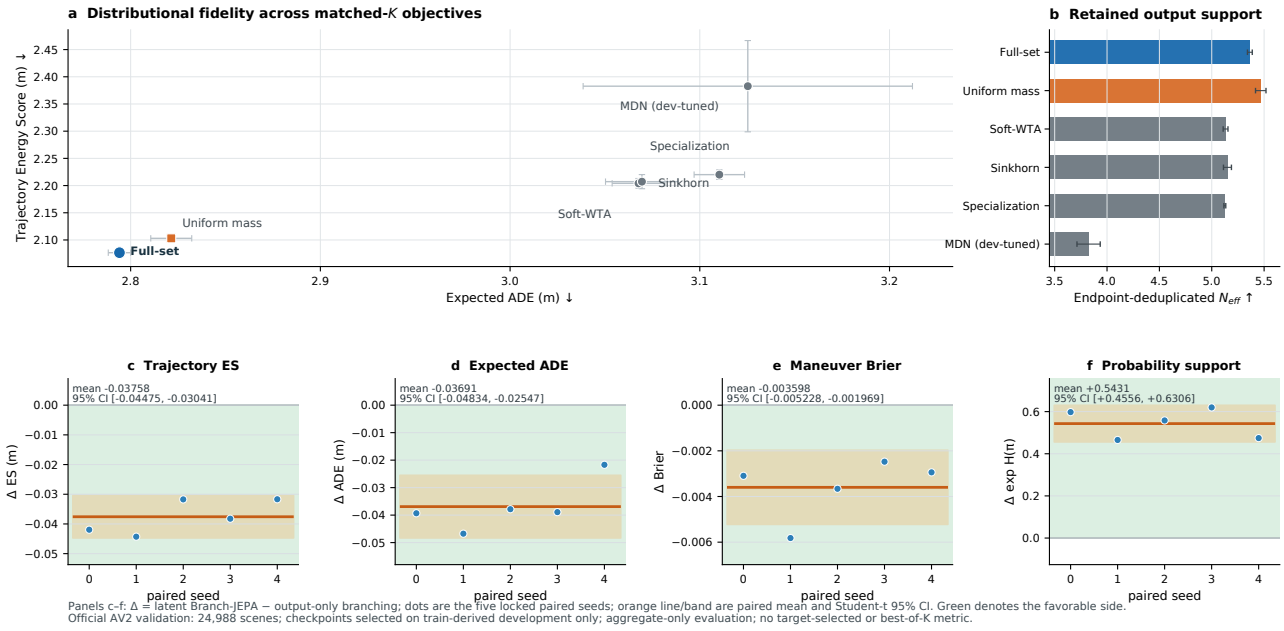


Figure S1: **Official-validation distribution and architecture summary.** Panels (a)–(b) compare all six locked, matched- $K=6$ objectives on the 24,988-scene official AV2 validation split. Lower and farther left is better in (a); (b) reports endpoint-deduplicated effective support rather than nominal head count. Panels (c)–(f) show every paired seed difference for latent Branch-JEPA minus the parameter-exact output-only control; dots are the five locked seeds and orange bands are paired- t 95% intervals. The two support panels use their explicitly labeled registered summaries: endpoint-deduplicated support for the objective matrix and probability support $\exp H(\pi)$ for the architecture isolation pair. Checkpoints were selected only on the disjoint train-derived development split. No panel uses a target-selected candidate or best-of- K metric.

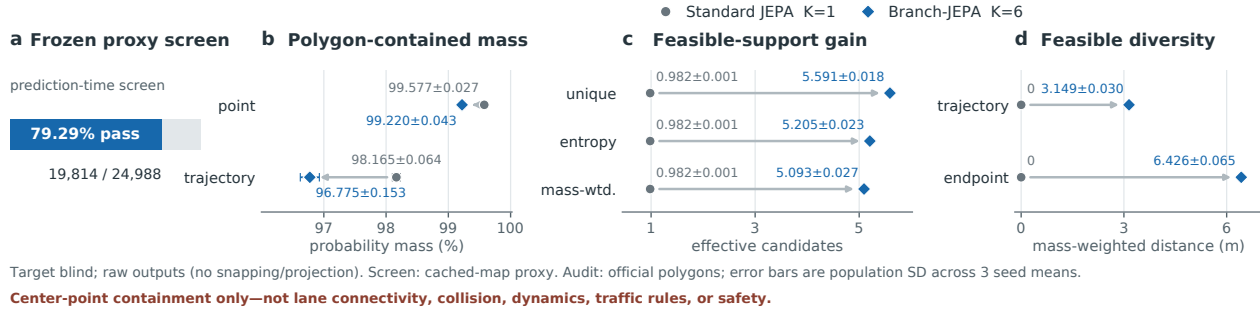


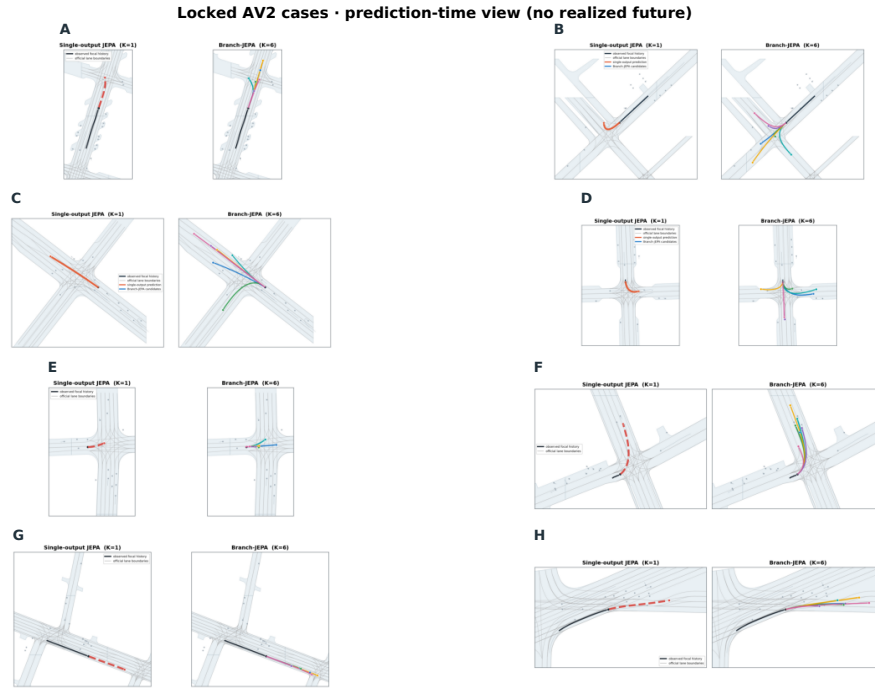
Figure S2: **Target-blind AV2 proxy and official raw-map audit.** (a) Frozen cached-map screen; (b) official-polygon point and trajectory mass; (c) feasible effective-support measures; and (d) feasible trajectory/endpoint diversity. Arrows run from Standard JEPA ($K=1$) to Branch-JEPA ($K=6$), and labels are mean±population SD across three seed means.

router mass is carried by candidates satisfying that threshold. This frozen target-blind screen passes 19,814/24,988 = 79.29% of validation contexts. The six displayed contexts satisfy the stricter conditions that every candidate has zero points beyond the 3 m proxy and that every raw candidate point is inside an official drivable polygon. Their identities, raw-map hashes, per-candidate containment values, and source-image hashes are fixed in the archived figure manifest before the realized future is added for reference. This case-level check and the population-level official-polygon audit

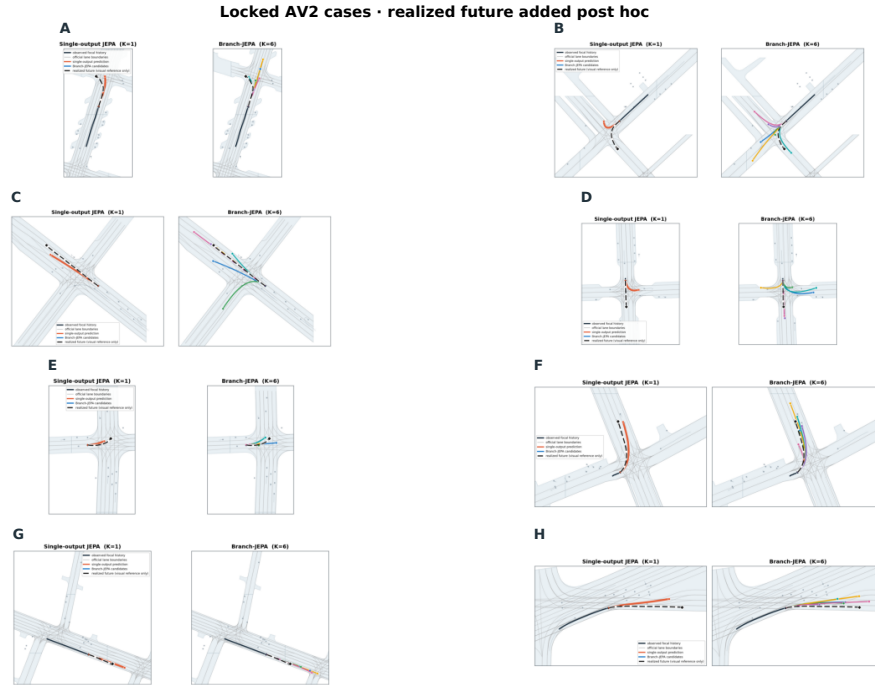
in Table S9 establish center-point containment only; neither guarantees lane connectivity nor that the displayed alternatives cover the realized future.

Extended AV2 prediction atlas

Figure S3 shows the same eight pre-existing target-blind locks twice in the same order. The top panel is exactly the prediction-time view: observed history, official map geometry, the single-output prediction, and all six raw Branch-JEPA candidates. The bottom panel changes only one layer



Eight pre-existing target-blind locks. All six raw Branch-JEPA candidates and the single-output baseline are fully inside official drivable polygons; no projection, snapping, filtering, or candidate removal.



Same eight cases as the no-GT atlas. The dashed realized future was loaded only after the target-blind lock and is a visual reference, not a selection signal.

Figure S3: **Paired locked AV2 atlas.** Top: the prediction-time view of all eight pre-existing target-blind locks, containing the raw single path and all six raw Branch-JEPA paths. Bottom: the identical cases, maps, and predictions with only the dashed realized future added after locking. Every displayed trajectory has 100% center-point containment in official drivable polygons; the bottom panel is a post-hoc diagnostic, not a selection signal.

by adding the realized future after the lock. Thus a reader can first inspect the model’s enumerated alternatives without outcome information and then use the lower panel as a post-hoc diagnostic. Every displayed raw trajectory has 100% center-point containment in the official drivable polygons; none was snapped, projected, filtered, removed, or selected by target error.

Atlas audit key. The upper panel contains prediction-time information only; the lower panel adds the future after locking. Both retain the unchanged single path and all six Branch-JEPA paths for every case.

S10 Cross-Domain Interface Diagnostics

The following experiments evaluate Branch-JEPA with specialization training. They test whether the shared multi-branch architecture exposes context-dependent successor support across observation regimes; the AV2 experiments separately compare specialization with full-set training.

OGBench information and readout protocols. Table S10 combines distinct, artifact-backed campaigns and is not a set of measurements of one common checkpoint set. The planAll campaign fits action-marginalized point-maze predictors to current coordinates; its learned arms and separately fitted codebook control appear in panel (a). A second five-seed retraining campaign (valid2) contains only Standard JEPA, MDN, and Branch-JEPA and supplies both verified-route existence and transition precision in panel (b). Both graph campaigns use $K=8$, 30k updates, 512-dimensional latents, coordinate features, and the same edge-decoding rule, but their trained checkpoints differ.

For these graph diagnostics, 20% of unique directed edges are withheld from fitting subject to retaining source-cell training support. A candidate creates an edge when its cosine similarity to a cell latent exceeds .5; all heads are retained. planAll is the success rate on up to 300 sampled, truly reachable start–goal queries under predicted-graph BFS, not exhaustive all-pairs evaluation. Verified-route existence uses a separate query set, retains only predicted edges that are real transitions in the full environment graph, and asks whether a complete path remains. Transition precision is the macro-average of per-source fractions of proposed edges that are real.

The higher-dimensional campaigns in panel (c) use still other checkpoints and report raw candidate coverage only. They are not interchangeable with the graph diagnostics, and none of these quantities is closed-loop task success.

Case-audit protocols. The point-maze tensor archives record emissions and context-only router masses before any realized successor is consulted. Archive A is an independently trained one-seed extraction whose five fixed contexts exactly match the checked-in producer’s cell list; Figure S5 (a) shows all five rather than selecting favorable rows. Archive B, used in Figure S4(a), contains all four contexts from a second extraction. The two archives are mechanism snapshots, not repeated seeds for a common aggregate.

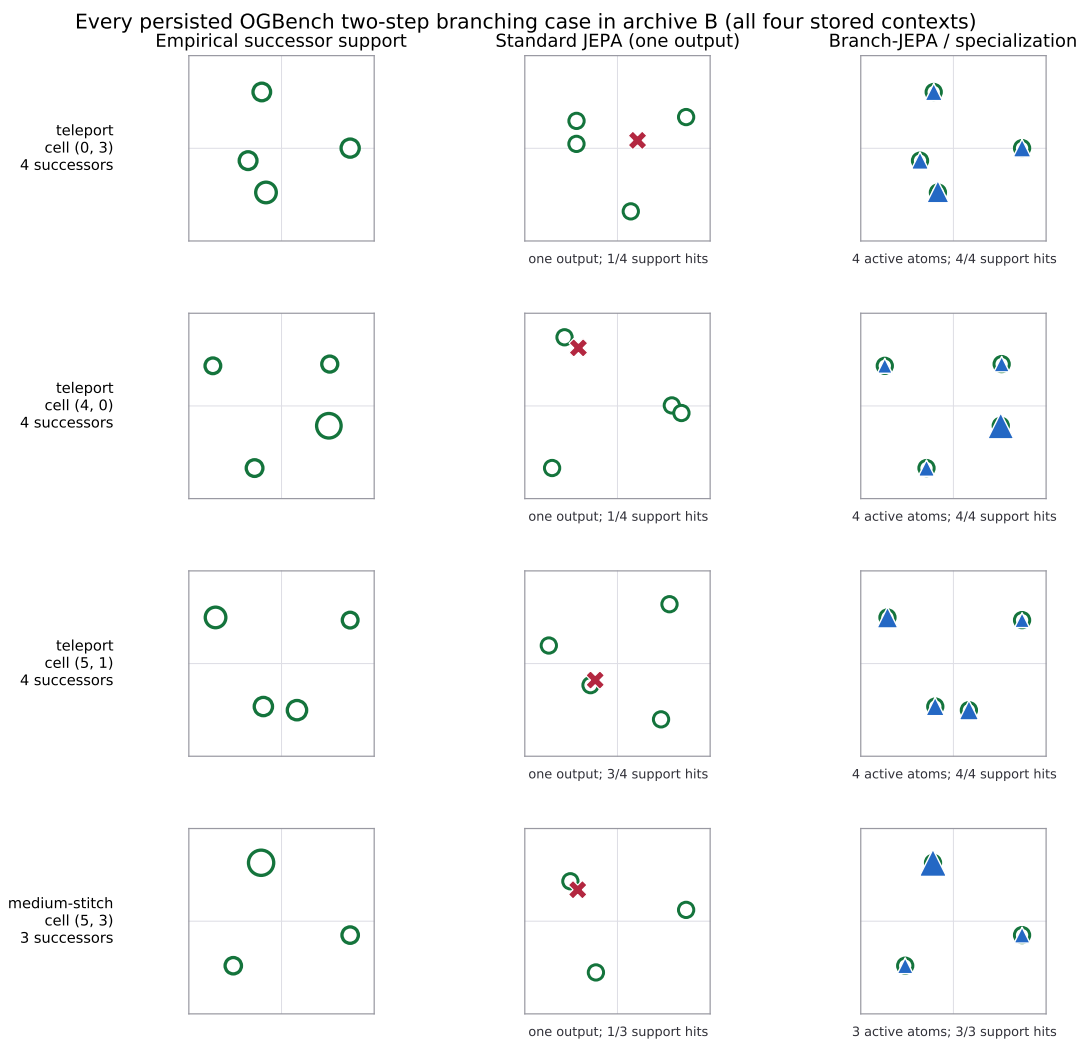
For the episode-held-out AntMaze audit, a separate one-seed campaign splits 1,000 complete teleport episodes into

700 training and 300 held-out episodes using a fixed split seed. At stride 25 and horizon 8, all 110 eligible evaluation contexts come only from held-out episodes. Figure S5(b) uses context index zero in the frozen deterministic ordering, not a context selected using future outcome, model error, or method margin. Metrics aggregate every retained branch with its context-only router mass. This campaign is separate from the three-seed raw-COV-R campaign in Table S10.

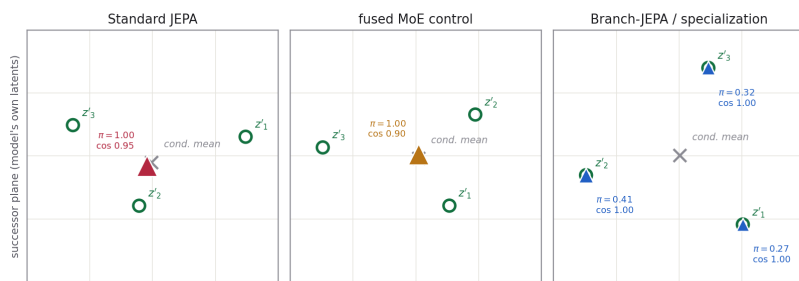
How to read the OGBench atlases. Across all nine archived entries, green circles are empirical successors added after prediction, red crosses are Standard-JEPA outputs, and blue triangles are all router-active Branch-JEPA atoms; marker area encodes empirical frequency or context-only router mass, and a support hit uses the fixed .5 cosine threshold.

Figure inputs (SHA-256 prefixes) are `head_specialization.json` (4ec6adc3) and `branching_complexity.json` (d4b7322f).

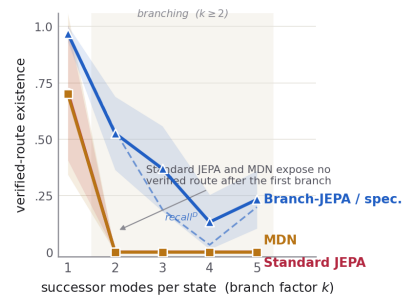
A separate three-seed execution control is inconclusive: medium-stitch replanning is $86.7 \pm 11.5 / 66.7 \pm 11.5\%$ for MDN/Branch-JEPA (mean $\pm$ sample SD); large-stitch and a one-seed DINO-WM pilot are also inconclusive.



(a) Every context in two-step archive B



(b) Trained successor geometry



(c) Branch-factor stress test

Figure S4: **OGBench finite-support cases and specialization diagnostics.** (a) All four immutable archive-B contexts. (b) At a three-successor state, three active atoms align with distinct successor latents; five-seed coverage is $76.1 \pm 23.9\%$, versus $33.3 \pm 0.1\% / 32.8 \pm 0.5\%$ for Standard/fused readouts. (c) Held-out verified-route existence under increasing branch factor; $k=4/5$ remain unresolved at $13.4 \pm 13.5\% / 23.3 \pm 14.3\%$.

Arm	Seed	ES-traj.	ES-end.	Brier	ECE	traj. observation	exp $H(\pi)$	active
Point $K=1$	0	2.8387	5.5717	.11942	.05971	2.8387	1.000	1
	1	2.8449	5.5842	.12358	.06179	2.8449	1.000	1
	2	2.8493	5.5892	.12078	.06039	2.8493	1.000	1
	3	2.8419	5.5742	.11998	.05999	2.8419	1.000	1
	4	2.8381	5.5647	.12102	.06051	2.8381	1.000	1
Branch-JEPA / full-set, learned π	0	2.0755	4.0647	.09224	.01644	3.6217	5.779	6
	1	2.0723	4.0479	.09030	.01309	3.6137	5.757	6
	2	2.0760	4.0580	.09198	.01832	3.6246	5.813	6
	3	2.0728	4.0575	.09317	.01774	3.6074	5.851	6
	4	2.0864	4.0731	.09200	.01346	3.6251	5.786	6
Branch-JEPA / full-set, uniform π	0	2.1083	4.1182	.09314	.01485	3.6605	6.000	6
	1	2.0996	4.1056	.09248	.01499	3.6270	6.000	6
	2	2.1064	4.1214	.09417	.01387	3.6626	6.000	6
	3	2.0969	4.1063	.09410	.01545	3.6428	6.000	6
	4	2.1036	4.1143	.09191	.01472	3.6483	6.000	6
Branch-JEPA / soft-WTA	0	2.1938	4.3041	.09314	.01272	3.9726	5.537	6
	1	2.2083	4.3285	.09319	.01508	3.9953	5.509	6
	2	2.2124	4.3429	.09431	.01499	3.9968	5.537	6
	3	2.2110	4.3455	.09386	.01364	4.0077	5.526	6
	4	2.1952	4.3058	.09311	.01220	3.9658	5.529	6
Branch-JEPA / partial Sinkhorn	0	2.1910	4.3033	.09338	.01357	3.9612	5.505	6
	1	2.2048	4.3233	.09365	.01336	3.9867	5.466	6
	2	2.2236	4.3611	.09458	.01290	4.0178	5.474	6
	3	2.2165	4.3501	.09367	.01245	4.0041	5.498	6
	4	2.2000	4.3196	.09410	.01355	3.9753	5.503	6
Branch-JEPA / specialization	0	2.2163	4.3430	.09333	.01242	4.0315	5.325	6
	1	2.2140	4.3385	.09293	.01321	4.0340	5.288	6
	2	2.2153	4.3413	.09350	.01169	4.0243	5.316	6
	3	2.2355	4.3858	.09445	.01260	4.0699	5.312	6
	4	2.2195	4.3521	.09380	.01248	4.0380	5.326	6
MDN / development-tuned	0	2.3438	4.6042	.09972	.02545	3.9742	3.783	6
	1	2.5272	4.9753	.10683	.03201	4.2579	3.810	6
	2	2.3157	4.5469	.09883	.02036	4.0460	4.064	6
	3	2.3475	4.6108	.10024	.02454	3.9967	3.841	6
	4	2.3796	4.6775	.10086	.02517	4.0275	3.775	6

Matched- K aggregate expected and top-1 point metrics

Arm	expected ADE	expected FDE	top-1 ADE	top-1 FDE	top-1 MR
Branch-JEPA / full-set	2.7941 $\pm$.0059	7.1904 $\pm$.0170	2.4694 $\pm$.0888	6.2939 $\pm$.2270	.7236 $\pm$.0166
Branch-JEPA / full-set, uniform	2.8214 $\pm$.0108	7.2293 $\pm$.0410	2.7383 $\pm$.4421	7.0138 $\pm$ 1.2861	.7795 $\pm$.0608
Branch-JEPA / soft-WTA	3.0677 $\pm$.0139	7.9442 $\pm$.0347	2.4858 $\pm$.0223	6.3828 $\pm$.0564	.7014 $\pm$.0018
Branch-JEPA / partial Sinkhorn	3.0695 $\pm$.0192	7.9448 $\pm$.0363	2.5143 $\pm$.0117	6.4523 $\pm$.0322	.7056 $\pm$.0018
Branch-JEPA / specialization	3.1103 $\pm$.0133	8.0348 $\pm$.0360	2.4932 $\pm$.0125	6.3876 $\pm$.0350	.7018 $\pm$.0016
MDN / development-tuned	3.1253 $\pm$.0868	8.0942 $\pm$.2273	2.5609 $\pm$.0860	6.5957 $\pm$.2245	.7179 $\pm$.0084

Paired full-set minus specialization statistics

Metric	mean difference	paired- t 95% interval	common direction
Trajectory ES	-.1435	[-.1574, -.1295]	5/5 seeds
Trajectory observation	-.4211	[-.4512, -.3909]	5/5 seeds
Expected ADE (m)	-.3161	[-.3392, -.2930]	5/5 seeds

Table S4: Every locked seed on the 24,988-scene official AV2 validation split, plus the parameter-matched $K=1$ structural control, followed by aggregate expected and router-top-1 point metrics for the six $K=6$ rows. Distances are in meters. The trajectory observation term is ES-traj plus one half of probability-weighted pairwise trajectory diversity. Every configuration and checkpoint was selected on the disjoint train-derived development split before this one-shot evaluation; no official-split retuning, seed removal, or checkpoint replacement was performed.

Branch location	Seed	ES-traj. ↓	expected ADE ↓	Brier ↓	exp $H(\pi)$ ↑	top-1 ADE ↓
Latent (Branch-JEPA)	0	2.07554	2.79683	.092245	5.7790	2.44859
	1	2.07235	2.79085	.090297	5.7571	2.36108
	2	2.07605	2.79954	.091981	5.8128	2.60835
	3	2.07281	2.78544	.093169	5.8512	2.46677
	4	2.08639	2.79807	.092000	5.7864	2.46198
Output only (one latent)	0	2.11748	2.83616	.095338	5.1814	2.46605
	1	2.11664	2.83760	.096117	5.2917	2.44981
	2	2.10778	2.83739	.095643	5.2547	2.43477
	3	2.11105	2.82435	.095644	5.2312	2.41962
	4	2.11807	2.81976	.094942	5.3120	2.46466
Latent mean±SD	—	2.07663±.00569	2.79414±.00588	.091938±.001038	5.7973±.0361	2.4694±.0888
Output mean±SD	—	2.11421±.00455	2.83105±.00839	.095537±.000434	5.2542±.0514	2.4470±.0199

Table S5: **Latent branching versus output-only branching on official validation.** Both emit six weighted trajectories, use exactly 3,608,958 downstream trainable parameters, the same five seeds and 30-epoch budget, and target-blind inference. Checkpoints were selected on the train-derived development split and evaluated once without official-split tuning. Branch-JEPA improves ES, expected ADE, Brier, and mass-effective support in all five paired seeds. The top-1 ADE difference is unresolved; this control isolates finite-support representation rather than a universal point-error gain.

control	λ_z	λ_y	λ_{rec}	target signal	role	ES-traj. ↓	expected ADE ↓	Brier ↓	exp $H(\pi)$ ↑
Decoded set only	0	1	0	no	target-path ablation	2.05772±.00819	2.79041±.01284	.09465±.00163	5.593±.042
Latent + decoded	1	1	0	yes	reconstruction ablation	2.04925±.01067	2.76141±.01288	.09354±.00080	5.789±.017
Full objective	1	1	1	yes	paper setting	2.05603±.00329	2.76438±.01326	.09328±.00141	5.806±.010
$\lambda_z = .5$	.5	1	1	yes	latent-weight sweep	2.05930±.00707	2.78107±.01974	.09431±.00104	5.799±.038
$\lambda_z = 2$	2	1	1	yes	latent-weight sweep	2.05395±.01221	2.76396±.02129	.09269±.00139	5.771±.035
$\lambda_y = .5$	1	.5	1	yes	decoded-weight sweep	2.05765±.01104	2.76998±.01496	.09335±.00071	5.760±.055
$\lambda_y = 2$	1	2	1	yes	decoded-weight sweep	2.05459±.00530	2.77643±.01912	.09403±.00132	5.766±.018

Table S6: **Five-seed loss and weight ablation on the train-derived development split.** All rows keep the $K=6$ latent-branch architecture, initialization protocol, and 30-epoch budget. The decoded-only row removes all target-path gradients, although the inert EMA tower remains instantiated. Relative to decoded-only, the full objective improves expected ADE and effective support in every paired seed; the corresponding full-minus-control 95% intervals are $[-.0476, -.0044]$ and $[.155, .271]$. Trajectory-ES and Brier intervals overlap zero. The weight sweep shows that the result is not sensitive to a narrow λ_z/λ_y choice.

Arm	Seed	latent ES	traj. ES (m)	end. ES (m)	Brier	ECE	top-1 FDE (m)	$N_{eff}^{.10}$
Point $K=1$	0	.21030	2.8843	5.6710	.12142	.06071	5.6710	1.000
Point $K=1$	1	.20228	2.8769	5.6540	.12230	.06115	5.6540	1.000
Point $K=1$	2	.21152	2.9147	5.7258	.12142	.06071	5.7258	1.000
Branch-JEPA / full-set $K=6$	0	.15431	2.1286	4.1778	.09387	.01557	6.3528	4.241
Branch-JEPA / full-set $K=6$	1	.14838	2.1171	4.1459	.09074	.01232	6.2698	4.220
Branch-JEPA / full-set $K=6$	2	.15469	2.1454	4.2112	.09329	.01429	6.4211	4.319

Table S7: Per-seed frozen-anchor Stage-B results. Target encoder and decoder drift are exactly zero in every run. The $K=1$ rows have lower top-1 FDE in every seed, while the Branch-JEPA rows improve full-distribution scores and retain collision-adjusted support above four; these metrics answer different prediction questions.

Seed	ADE	FDE	mean std	low-std frac.	eff. rank	SHA-256 prefix
0	.1827	.3216	.02187	0	12.911	7721aad75852
1	.1781	.2918	.02108	0	12.860	8585e6e7b7cf
2	.1841	.3192	.02184	0	13.196	a797ffe511bb

Table S8: Accepted Stage-A anchors. All selected checkpoints occur at epoch 30. Passing these numerical gates rules out the measured collapse modes but does not prove that the representation preserves every semantic distinction.

Metric	Standard JEPA $K=1$	Branch-JEPA / full-set $K=6$	Branch-Standard [95% CI]
Drivable point mass (%)	99.577 $\pm$.027	99.220 $\pm$.043	-.358 [-.386, -.329]
Drivable trajectory mass (%)	98.165 $\pm$.064	96.775 $\pm$.153	-1.390 [-1.493, -1.286]
Feasible unique support	.982 $\pm$.001	5.591 $\pm$.018	+4.609 [4.600, 4.617]
Feasible entropy support (empty=0)	.982 $\pm$.001	5.205 $\pm$.023	+4.223 [4.212, 4.233]
Mass-weighted feasible effective support	.982 $\pm$.001	5.093 $\pm$.027	+4.111 [4.098, 4.123]
Mass-weighted feasible trajectory diversity (m)	0	3.149 $\pm$.030	+3.149 [3.133, 3.164]
Mass-weighted feasible endpoint diversity (m)	0	6.426 $\pm$.065	+6.426 [6.393, 6.458]

Table S9: Official-map audit (mean and population SD across three seed means). Intervals are paired-scene-bootstrap intervals conditional on those fitted models.

(a) planAll: independent graph-fit campaign (%)

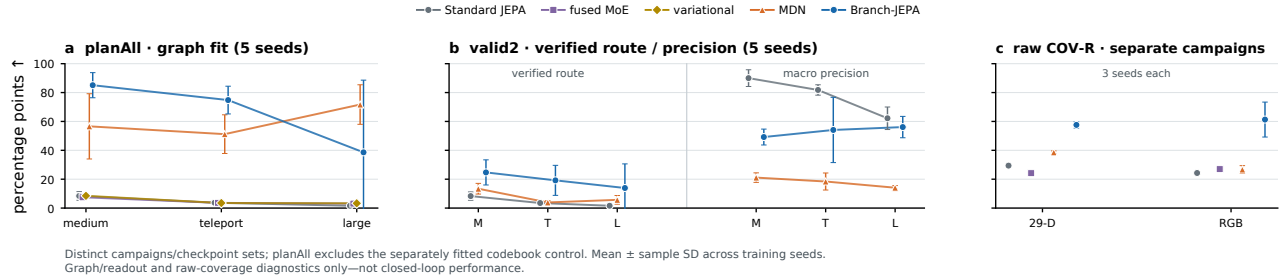
predictor	medium	teleport	large
Standard JEPA	8.4 $\pm$ 3.0	3.7 $\pm$ 1.0	1.6 $\pm$ 0.7
fused MoE control	7.5 $\pm$ 1.7	3.6 $\pm$ 0.9	3.1 $\pm$ 1.2
variational control	8.5 $\pm$ 1.6	3.5 $\pm$ 1.6	3.3 $\pm$ 0.4
MDN	56.6 $\pm$ 22.6	51.2 $\pm$ 13.4	71.7$\pm$13.7
Branch-JEPA	85.1$\pm$8.7	74.8$\pm$9.6	38.6 $\pm$ 50.0
codebook control [†]	26.7 $\pm$ 2.7	16.7 $\pm$ 1.9	14.9 $\pm$ 4.0

(b) valid2: verified route and macro transition precision (%)

predictor	verified route $\uparrow$			macro transition precision $\uparrow$		
	medium	teleport	large	medium	teleport	large
Standard JEPA	8.3 $\pm$ 2.9	3.5 $\pm$ 0.9	1.6 $\pm$ 0.7	90.0 $\pm$ 5.8	81.8 $\pm$ 3.6	62.2 $\pm$ 7.8
MDN	13.4 $\pm$ 3.7	3.9 $\pm$ 1.2	5.7 $\pm$ 3.0	21.1 $\pm$ 3.3	18.4 $\pm$ 5.9	14.1 $\pm$ 1.5
Branch-JEPA	24.7$\pm$8.7	19.2$\pm$10.4	13.9$\pm$16.7	49.2 $\pm$ 5.5	54.1 $\pm$ 22.6	56.1 $\pm$ 7.4

(c) Separate higher-dimensional raw-support campaigns (%)

observation / metric $\uparrow$	Standard	fused	MDN	Branch-JEPA
29-D AntMaze / raw COV-R	29.4 $\pm$ 1.0	24.2 $\pm$ 0.4	38.6 $\pm$ 1.0	57.6$\pm$2.3
RGB AntMaze / raw COV-R	24.3 $\pm$ 1.2	27.0 $\pm$ 0.7	26.7 $\pm$ 2.7	61.3$\pm$12.1



(d) Campaign-separated visual summary

Table S10: **Campaign-separated OGBench diagnostics.** Panels (a)–(c) report the independent graph-fit, valid2, and higher-dimensional campaigns; (d) visualizes the same campaign-separated aggregates. The [†]codebook is separately fitted, and all values are mean $\pm$ sample SD across training seeds in percentage points. Campaigns do not share checkpoints.

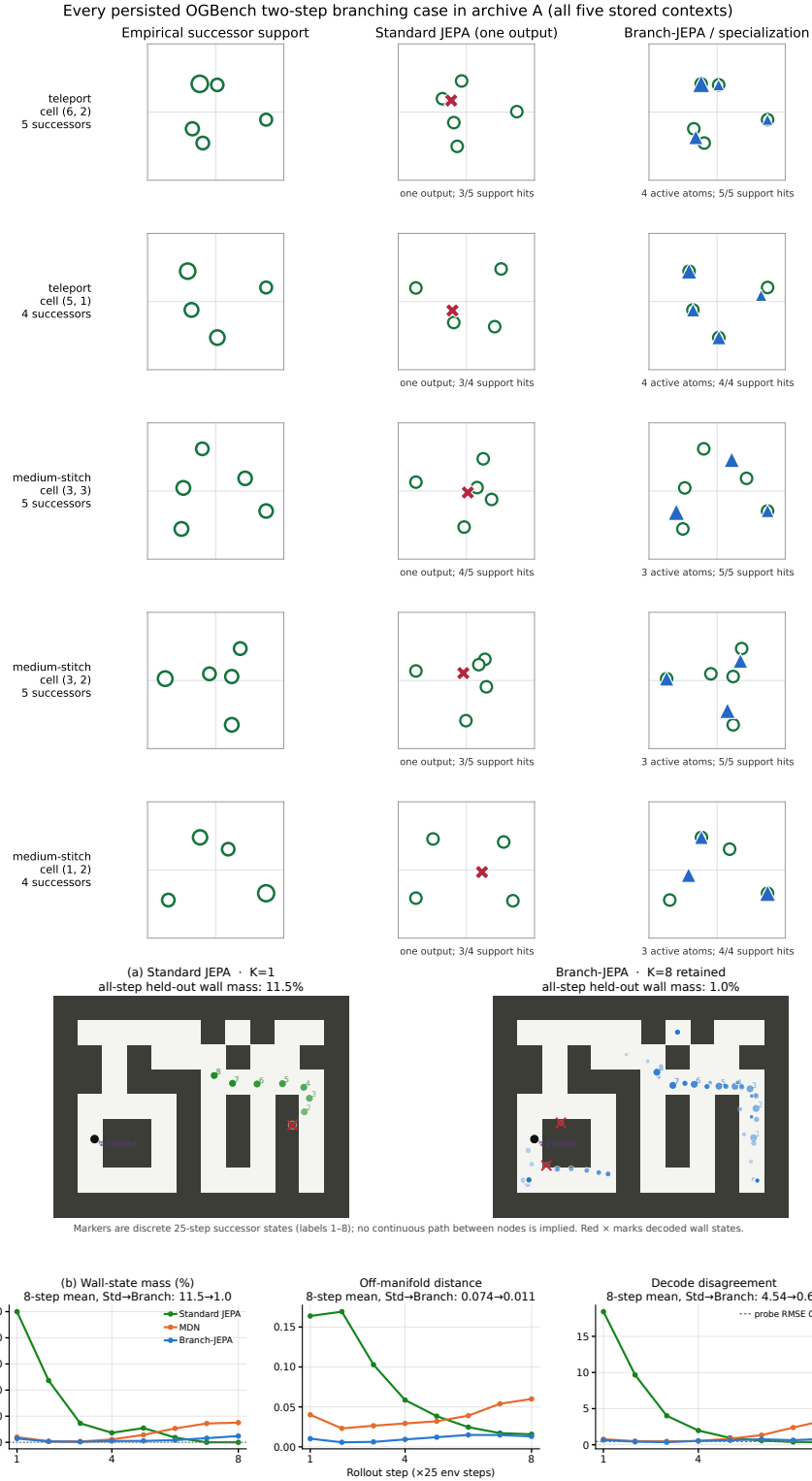


Figure S5: **Additional target-blind support audits.** (a) All five fixed archive-A contexts: empirical successors, Standard JEPA’s single prediction, and every router-active Branch-JEPA atom. Empirical support is added only after emission for this post-hoc projection. (b) The first frozen held-out AntMaze context and aggregate curves over all 110 eligible contexts. Red crosses mark probe-decoded wall cells; nodes are 25 environment steps apart and do not imply a continuous collision-free path.